

# Sid Mathur's Thematic ETF Screener – Daily Report

2026-08-27

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## Action Center

**Run date:** 2026-08-27    **Universe:** 659 ETFs / 26 clusters

**Full PC1:** LONG since 2026-08-04 (D+16, 0.3%)    **Clusters long:** 11/26    **ETFs long:** 335/659 (51%)

### Today's events (cluster level)

spread\_buy = 5dma – 50dma; spread\_sell = 5dma – 20dma (price units). A BUY fired today iff spread\_buy crossed above 0; a SELL iff spread\_sell crossed below the cushioned threshold.

#### SELL (2)

| cluster   | theme                 | spread_sell | p_h5 | nss    |
|-----------|-----------------------|-------------|------|--------|
| cluster_6 | U.S. Large Cap Growth | -2.540      | 0.62 | -0.151 |
| cluster_1 | U.S. Large Cap Equity | -0.805      | 0.59 | 0.303  |

### Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p\_h5 is the 5-day probability for context. **spread\_buy = 5dma – 50dma, spread\_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

#### Imminent BUY (top 4 by h=1 NSS, filter-approved)

| cluster    | theme                | spread_buy | p_h1 | p_h5 | nss_h1 | nss_h5 |
|------------|----------------------|------------|------|------|--------|--------|
| cluster_10 | Healthcare           | 7.783      | 1    | 1.00 | 1.000  | 1.00   |
| cluster_8  | U.S. Dividend Equity | 4.228      | 1    | 1.00 | 1.000  | 0.99   |
| cluster_4  | U.S. Large Cap Value | 2.692      | 1    | 1.00 | 0.603  | 0.50   |
| cluster_13 | Financials           | 2.719      | 1    | 0.99 | 0.072  | 0.15   |

#### Imminent SELL (top 1 by h=1 NSS)

| cluster    | theme            | spread_sell | p_h1  | p_h5 | nss_h1 | nss_h5 |
|------------|------------------|-------------|-------|------|--------|--------|
| cluster_19 | Consumer Staples | 0.164       | 0.333 | 0.52 | -0.149 | -0.4   |

**Full Sample PC1**

Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

**Replication basket:** VOO basket cor: 0.94 sticky

**Current state:** LONG since 2026-08-04 (16 days, 0.3%)

5dma: 145.438 20dma: 145.100 50dma: 143.206

Spread vs 50dma: +2.232 Spread vs 20dma: +0.338

P(buy crossover within 5 days): 100.0% P(sell): 22.9%

## Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

### Active positions (long clusters)

| ticker | cluster    | theme                          | nss    | days | return | pos_wt | cor  |
|--------|------------|--------------------------------|--------|------|--------|--------|------|
| SRLN   | cluster_23 | Bank Loan / Floating Rate      | 1.000  | 7    | 0.2%   | 6.01%  | 0.89 |
| FLOT   | cluster_23 | Bank Loan / Floating Rate      | 1.000  | 7    | 0.2%   | 5.88%  | 0.89 |
| GUNR   | cluster_11 | Materials                      | 1.000  | 15   | 5.5%   | 5.53%  | 0.98 |
| BKLN   | cluster_23 | Bank Loan / Floating Rate      | 1.000  | 7    | 0.2%   | 4.30%  | 0.89 |
| FLRN   | cluster_23 | Bank Loan / Floating Rate      | 1.000  | 7    | 0.2%   | 2.46%  | 0.89 |
| IAU    | cluster_17 | Gold                           | 1.000  | 13   | 8.6%   | 2.29%  | 0.95 |
| SLV    | cluster_17 | Gold                           | 1.000  | 13   | 8.6%   | 1.25%  | 0.95 |
| AML    | cluster_11 | Materials                      | 1.000  | 15   | 5.5%   | 0.99%  | 0.98 |
| URA    | cluster_11 | Materials                      | 1.000  | 15   | 5.5%   | 0.69%  | 0.98 |
| FLTR   | cluster_23 | Bank Loan / Floating Rate      | 1.000  | 7    | 0.2%   | 0.68%  | 0.89 |
| GLD    | cluster_17 | Gold                           | 1.000  | 13   | 8.6%   | 0.07%  | 0.95 |
| IEMG   | cluster_7  | Emerging Markets Equity        | 0.996  | 10   | 1.3%   | 4.67%  | 0.99 |
| ICSH   | cluster_24 | Cash / Ultra-Short Fixed In... | 0.958  | 7    | 0.1%   | 7.97%  | 0.86 |
| SHV    | cluster_24 | Cash / Ultra-Short Fixed In... | 0.958  | 7    | 0.1%   | 5.08%  | 0.86 |
| MINT   | cluster_24 | Cash / Ultra-Short Fixed In... | 0.958  | 7    | 0.1%   | 3.47%  | 0.86 |
| GBIL   | cluster_24 | Cash / Ultra-Short Fixed In... | 0.958  | 7    | 0.1%   | 2.51%  | 0.86 |
| PDBC   | cluster_22 | Broad Commodities              | 0.942  | 12   | 1.9%   | 5.15%  | 0.98 |
| VEA    | cluster_2  | Developed ex-US Equity         | 0.931  | 19   | 2.1%   | 8.73%  | 0.99 |
| IGV    | cluster_15 | Technology                     | 0.912  | 16   | 1.2%   | 4.79%  | 0.90 |
| EWJ    | cluster_21 | Japan Equity                   | 0.437  | 14   | 0.6%   | 5.25%  | 0.97 |
| XLP    | cluster_19 | Consumer Staples               | -0.401 | 1    | -0.1%  | 5.65%  | 0.99 |
| XLU    | cluster_19 | Consumer Staples               | -0.401 | 1    | -0.1%  | 3.49%  | 0.99 |
| VNQ    | cluster_14 | Real Estate                    | -0.445 | 1    | -0.6%  | 8.29%  | 0.99 |

### Watchlist (imminent-buy clusters, $p > 80\%$ , not yet long)

| ticker | cluster    | theme                | p_h5 | nss   | weight | cor  |
|--------|------------|----------------------|------|-------|--------|------|
| VTV    | cluster_4  | U.S. Large Cap Value | 1.00 | 0.504 | 1.092  | 0.97 |
| VHT    | cluster_10 | Healthcare           | 1.00 | 1.000 | 1.088  | 0.93 |
| SCHD   | cluster_8  | U.S. Dividend Equity | 1.00 | 0.991 | 0.891  | 0.96 |

|      |            |                            |      |       |       |      |
|------|------------|----------------------------|------|-------|-------|------|
| BWX  | cluster_25 | International Fixed Income | 1.00 | 0.983 | 0.752 | 0.93 |
| INDA | cluster_18 | India Equity               | 1.00 | 0.835 | 0.641 | 0.91 |
| NOBL | cluster_18 | India Equity               | 1.00 | 0.835 | 0.078 | 0.91 |
| PPLT | cluster_18 | India Equity               | 1.00 | 0.835 | 0.029 | 0.91 |
| XLF  | cluster_13 | Financials                 | 0.99 | 0.149 | 1.094 | 0.96 |
| V00  | cluster_1  | U.S. Large Cap Equity      | 0.89 | 0.303 | 1.007 | 0.99 |

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## Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

|    | id         | name                           | category             | n  | signal | change | days | return | nss    | dNSS   |
|----|------------|--------------------------------|----------------------|----|--------|--------|------|--------|--------|--------|
| 10 | cluster_10 | Healthcare                     | Healthcare           | 17 | FLAT   |        |      |        | 1.000  | -0.000 |
| 11 | cluster_11 | Materials                      | Materials            | 17 | LONG   |        | 15   | 5.5%   | 1.000  | +0.000 |
| 17 | cluster_17 | Gold                           | Gold                 | 12 | LONG   |        | 13   | 8.6%   | 1.000  | -0.000 |
| 23 | cluster_23 | Bank Loan / Floating Rate      | Bank Loan / Float... | 9  | LONG   |        | 7    | 0.2%   | 1.000  | +0.000 |
| 7  | cluster_7  | Emerging Markets Equity        | Emerging Markets ... | 30 | LONG   |        | 10   | 1.3%   | 0.996  | +0.010 |
| 8  | cluster_8  | U.S. Dividend Equity           | U.S. Dividend Equ... | 25 | FLAT   |        |      |        | 0.991  | -0.004 |
| 25 | cluster_25 | International Fixed Income     | International Fix... | 6  | FLAT   |        |      |        | 0.983  | -0.015 |
| 24 | cluster_24 | Cash / Ultra-Short Fixed In... | Cash / Ultra-Shor... | 9  | LONG   |        | 7    | 0.1%   | 0.958  | +0.006 |
| 22 | cluster_22 | Broad Commodities              | Broad Commodities    | 9  | LONG   |        | 12   | 1.9%   | 0.942  | -0.009 |
| 2  | cluster_2  | Developed ex-US Equity         | Developed ex-US E... | 79 | LONG   |        | 19   | 2.1%   | 0.931  | -0.014 |
| 15 | cluster_15 | Technology                     | Technology           | 14 | LONG   |        | 16   | 1.2%   | 0.912  | -0.034 |
| 18 | cluster_18 | India Equity                   | India Equity         | 12 | FLAT   |        |      |        | 0.835  | +0.010 |
| 4  | cluster_4  | U.S. Large Cap Value           | U.S. Large Cap Va... | 60 | FLAT   |        |      |        | 0.504  | -0.161 |
| 21 | cluster_21 | Japan Equity                   | Japan Equity         | 9  | LONG   |        | 14   | 0.6%   | 0.437  | +0.125 |
| 9  | cluster_9  | High Yield Bond                | High Yield Bond      | 23 | FLAT   |        |      |        | 0.392  | +0.143 |
| 1  | cluster_1  | U.S. Large Cap Equity          | U.S. Large Cap Eq... | 81 | FLAT   | →FLAT  |      |        | 0.303  | -0.123 |
| 13 | cluster_13 | Financials                     | Financials           | 14 | FLAT   |        |      |        | 0.149  | +0.043 |
| 20 | cluster_20 | Inflation-Protected Fixed I... | Inflation-Protect... | 10 | FLAT   |        |      |        | 0.055  | +0.030 |
| 5  | cluster_5  | Aggregate Fixed Income         | Aggregate Fixed I... | 58 | FLAT   |        |      |        | -0.006 | -0.002 |
| 6  | cluster_6  | U.S. Large Cap Growth          | U.S. Large Cap Gr... | 38 | FLAT   | →FLAT  |      |        | -0.151 | -0.021 |
| 12 | cluster_12 | China Equity                   | China Equity         | 15 | FLAT   |        |      |        | -0.186 | -0.128 |
| 3  | cluster_3  | U.S. Mid Cap Equity            | U.S. Mid Cap Equi... | 72 | FLAT   |        |      |        | -0.328 | -0.159 |
| 19 | cluster_19 | Consumer Staples               | Consumer Staples     | 11 | LONG   |        | 1    | -0.1%  | -0.401 | -0.121 |
| 14 | cluster_14 | Real Estate                    | Real Estate          | 14 | LONG   |        | 1    | -0.6%  | -0.445 | +0.065 |
| 16 | cluster_16 | Municipal Bond                 | Municipal Bond       | 13 | FLAT   |        |      |        | -0.542 | -0.117 |
| 26 | cluster_26 | Leveraged / Inverse            | Leveraged / Inver... | 2  | FLAT   |        |      |        | -0.558 | +0.082 |

## Cluster Details

### Cluster 1 – U.S. Large Cap Equity

**Members:** 81   **Var explained:** 0.858   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-08-05 – filter **PASS** (252d at date = +20.1%)
- Latest sell\_cross: 2026-08-26
- Current 252d return: +17.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-26 more recent than last effective buy 2026-08-05)



#### Category breakdown

| category                 | n_members | abs_weight |
|--------------------------|-----------|------------|
| U.S. Large Cap Equity    | 9         | 0.1191     |
| U.S. Total Market Equity | 7         | 0.0930     |
| Global Equity            | 7         | 0.0910     |
| Technology               | 5         | 0.0601     |
| Consumer Discretionary   | 5         | 0.0587     |
| Asset Allocation         | 4         | 0.0483     |
| Industrials              | 4         | 0.0407     |
| U.S. Equity              | 3         | 0.0388     |

|                                |   |        |
|--------------------------------|---|--------|
| U.S. Mega Cap Equity           | 3 | 0.0387 |
| U.S. Multifactor Equity        | 3 | 0.0387 |
| U.S. Quality / Factor Equity   | 3 | 0.0382 |
| U.S. Mid Cap Growth            | 3 | 0.0381 |
| U.S. Momentum Equity           | 3 | 0.0344 |
| Communication Services         | 3 | 0.0336 |
| U.S. ESG Equity                | 2 | 0.0264 |
| Momentum Equity                | 2 | 0.0238 |
| Convertible Bond               | 2 | 0.0227 |
| Commodity                      | 1 | 0.0132 |
| Large Cap Value Equities       | 1 | 0.0131 |
| Large Cap Blend Equities       | 1 | 0.0131 |
| Dividend Equity                | 1 | 0.0128 |
| U.S. Equal Weight Equity       | 1 | 0.0126 |
| U.S. Dividend Equity           | 1 | 0.0125 |
| Growth Equity                  | 1 | 0.0123 |
| IPO Equity                     | 1 | 0.0118 |
| U.S. Mid Cap Momentum Equity   | 1 | 0.0117 |
| Alternatives                   | 1 | 0.0115 |
| U.S. Small Cap Momentum Equity | 1 | 0.0112 |
| International IPO Equity       | 1 | 0.0101 |
| Israel Equity                  | 1 | 0.0097 |

**Replication basket** – VOO (cor=0.99) basket lead at lag 1: -0.009

| ticker | rep_wt | name                 | category              | etf_signal |
|--------|--------|----------------------|-----------------------|------------|
| VOO    | 1.007  | Vanguard S&P 500 ETF | U.S. Large Cap Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 50 | IWV    | 0.0133 | iShares Russell 3000 Fund                  | U.S. Total Market Equity | LONG       |
| 51 | IYY    | 0.0133 | iShares Dow Jones U.S. ETF                 | U.S. Total Market Equity | LONG       |
| 57 | IWB    | 0.0133 | iShares Russell 1000 ETF                   | U.S. Large Cap Equity    | LONG       |
| 62 | VTHR   | 0.0133 | Vanguard Russell 3000 ETF                  | U.S. Total Market Equity | LONG       |
| 65 | VONE   | 0.0133 | Vanguard Russell 1000 ETF                  | U.S. Large Cap Equity    | LONG       |
| 69 | ITOT   | 0.0133 | iShares Core S&P Total U.S. Stock Marke... | U.S. Total Market Equity | LONG       |
| 71 | SCHB   | 0.0133 | Schwab U.S. Broad Market ETF               | U.S. Total Market Equity | LONG       |
| 72 | SCHX   | 0.0133 | Schwab U.S. Large-Cap ETF                  | U.S. Large Cap Equity    | LONG       |
| 75 | VTI    | 0.0133 | Vanguard Total Stock Market ETF            | U.S. Total Market Equity | LONG       |
| 5  | SPY    | 0.0132 | State Street SPDR S&P 500 ETF Trust        | U.S. Large Cap Equity    | LONG       |

|    |      |        |                                             |                             |      |
|----|------|--------|---------------------------------------------|-----------------------------|------|
| 12 | URTH | 0.0132 | iShares MSCI World ETF                      | Global Equity               | LONG |
| 37 | TILT | 0.0132 | FlexShares Morningstar US Market Factors... | U.S. Multifactor Equity     | FLAT |
| 46 | JHML | 0.0132 | John Hancock Multifactor Large Cap ETF      | U.S. Large Cap Equity       | LONG |
| 49 | SUSA | 0.0132 | iShares ESG Optimized MSCI USA ETF          | U.S. ESG Equity             | LONG |
| 54 | SPYX | 0.0132 | State Street SPDR S&P 500 Fossil Fuel R...  | U.S. Equity                 | LONG |
| 56 | ESGU | 0.0132 | iShares ESG Aware MSCI USA ETF              | U.S. ESG Equity             | LONG |
| 63 | GSLC | 0.0132 | Goldman Sachs ActiveBeta U.S. Large Cap...  | Commodity                   | LONG |
| 67 | VV   | 0.0132 | Vanguard Large-Cap ETF                      | U.S. Large Cap Equity       | LONG |
| 68 | ILCB | 0.0132 | iShares Morningstar Large-Cap ETF           | U.S. Large Cap Equity       | LONG |
| 70 | IVV  | 0.0132 | iShares Core S&P 500 ETF                    | U.S. Large Cap Equity       | LONG |
| 73 | SPTM | 0.0132 | State Street SPDR Portfolio S&P 1500 Co...  | U.S. Total Market Equity    | LONG |
| 74 | VOO  | 0.0132 | Vanguard S&P 500 ETF                        | U.S. Large Cap Equity       | LONG |
| 7  | ACWI | 0.0131 | iShares MSCI ACWI ETF                       | Global Equity               | LONG |
| 13 | CRBN | 0.0131 | iShares Low Carbon Optimized MSCI ACWI ...  | Global Equity               | LONG |
| 14 | SPGM | 0.0131 | State Street SPDR Portfolio MSCI Global...  | Global Equity               | LONG |
| 15 | VT   | 0.0131 | Vanguard Total World Stock Index ETF        | Global Equity               | LONG |
| 48 | DSI  | 0.0131 | iShares ESG MSCI KLD 400 ETF                | Large Cap Blend Equities    | LONG |
| 52 | LRGF | 0.0131 | iShares U.S. Equity Factor ETF              | U.S. Multifactor Equity     | LONG |
| 58 | IWL  | 0.0131 | iShares Russell Top 200 ETF                 | U.S. Equity                 | LONG |
| 64 | EPS  | 0.0131 | WisdomTree U.S. LargeCap Fund               | Large Cap Value Equities    | LONG |
| 66 | MGC  | 0.0131 | Vanguard Mega Cap ETF                       | U.S. Mega Cap Equity        | LONG |
| 8  | AOA  | 0.0130 | iShares Core 80/20 Aggressive Allocatio...  | Asset Allocation            | LONG |
| 53 | OEF  | 0.0129 | iShares S&P 100 Fund                        | U.S. Mega Cap Equity        | LONG |
| 59 | QUAL | 0.0129 | iShares MSCI USA Quality Factor ETF         | U.S. Quality / Factor Eq... | FLAT |
| 36 | IMCG | 0.0128 | iShares Morningstar Mid-Cap Growth ETF      | U.S. Mid Cap Growth         | LONG |
| 44 | QDF  | 0.0128 | FlexShares Quality Dividend Index Fund      | Dividend Equity             | FLAT |
| 45 | CATH | 0.0128 | Global X S&P 500 Catholic Values ETF        | Global Equity               | LONG |
| 35 | VOT  | 0.0127 | Vanguard Mid-Cap Growth ETF                 | U.S. Mid Cap Growth         | LONG |
| 43 | IOO  | 0.0127 | iShares Global 100 ETF                      | Global Equity               | FLAT |
| 55 | XLG  | 0.0127 | Invesco S&P 500 Top 50 ETF                  | U.S. Mega Cap Equity        | FLAT |
| 60 | QUS  | 0.0127 | State Street SPDR MSCI USA StrategicFac...  | U.S. Quality / Factor Eq... | FLAT |
| 11 | AOR  | 0.0126 | iShares Core 60/40 Balanced Allocation ...  | Asset Allocation            | LONG |
| 18 | QQEW | 0.0126 | First Trust Nasdaq-100 Select Equal Wei...  | U.S. Equal Weight Equity    | LONG |
| 19 | XT   | 0.0126 | iShares Future Exponential Technologies...  | Technology                  | LONG |
| 28 | IWP  | 0.0126 | iShares Russell Midcap Growth ETF           | U.S. Mid Cap Growth         | LONG |
| 61 | SPHQ | 0.0126 | Invesco S&P 500 Quality ETF                 | U.S. Quality / Factor Eq... | LONG |
| 21 | FTC  | 0.0125 | First Trust Large Cap Growth AlphaDEX F...  | Technology                  | FLAT |
| 42 | SPHB | 0.0125 | Invesco S&P 500 High Beta ETF               | U.S. Equity                 | LONG |
| 47 | DGRW | 0.0125 | WisdomTree U.S. Quality Dividend Growth...  | U.S. Dividend Equity        | LONG |
| 81 | OMFL | 0.0124 | Invesco Russell 1000 Dynamic Multifacto...  | U.S. Multifactor Equity     | FLAT |
| 26 | RPG  | 0.0123 | Invesco S&P 500 Pure Growth ETF             | Growth Equity               | LONG |
| 4  | ROBO | 0.0122 | ROBO Global Robotics and Automation Ind...  | Technology                  | FLAT |
| 24 | IYC  | 0.0122 | iShares U.S. Consumer Discretionary ETF     | Consumer Discretionary      | LONG |
| 22 | PDP  | 0.0121 | Invesco Dorsey Wright Momentum ETF          | Momentum Equity             | FLAT |
| 31 | VCR  | 0.0120 | Vanguard Consumer Discretion ETF            | Consumer Discretionary      | LONG |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 34 | FDIS | 0.0120 | Fidelity MSCI Consumer Discretionary In... | Consumer Discretionary      | LONG |
| 77 | VFMO | 0.0120 | Vanguard U.S. Momentum Factor ETF          | U.S. Momentum Equity        | LONG |
| 3  | BOTZ | 0.0119 | Global X Robotics & Artificial Intellig... | Technology                  | FLAT |
| 1  | XLY  | 0.0118 | State Street Consumer Discretionary Sel... | Consumer Discretionary      | LONG |
| 10 | AOM  | 0.0118 | iShares Core 40/60 Moderate Allocation ... | Asset Allocation            | LONG |
| 23 | FPX  | 0.0118 | First Trust US Equity Opportunities ETF    | IPO Equity                  | FLAT |
| 16 | FV   | 0.0117 | First Trust Dorsey Wright Focus 5 ETF      | Momentum Equity             | LONG |
| 27 | XMMO | 0.0117 | Invesco S&P MidCap Momentum ETF            | U.S. Mid Cap Momentum Eq... | FLAT |
| 25 | CWB  | 0.0116 | State Street SPDR Bloomberg Convertible... | Convertible Bond            | LONG |
| 6  | QAI  | 0.0115 | NYLI Hedge Multi-Strategy Tracker ETF      | Alternatives                | LONG |
| 30 | MTUM | 0.0115 | iShares MSCI USA Momentum Factor ETF       | U.S. Momentum Equity        | FLAT |
| 32 | VOX  | 0.0113 | Vanguard Communication Services ETF        | Communication Services      | LONG |
| 33 | FCOM | 0.0113 | Fidelity MSCI Communication Services In... | Communication Services      | LONG |
| 38 | DWAS | 0.0112 | Invesco Dorsey Wright SmallCap Momentum... | U.S. Small Cap Momentum ... | FLAT |
| 29 | ICVT | 0.0111 | iShares Convertible Bond ETF               | Convertible Bond            | LONG |
| 17 | PRNT | 0.0110 | 3D Printing (The) ETF                      | Technology                  | FLAT |
| 2  | XLC  | 0.0109 | State Street Communication Services Sel... | Communication Services      | LONG |
| 9  | AOK  | 0.0109 | iShares Core 30/70 Conservative Allocat... | Asset Allocation            | LONG |
| 80 | PRN  | 0.0109 | Invesco Dorsey Wright Industrials Momen... | Industrials                 | FLAT |
| 76 | QMOM | 0.0108 | Alpha Architect U.S. Quantitative Momen... | U.S. Momentum Equity        | FLAT |
| 79 | RTH  | 0.0107 | VanEck Retail ETF                          | Consumer Discretionary      | FLAT |
| 39 | PPA  | 0.0102 | Invesco Aerospace & Defense ETF            | Industrials                 | FLAT |
| 41 | XAR  | 0.0102 | State Street SPDR S&P Aerospace & Defen... | Industrials                 | FLAT |
| 20 | FPXI | 0.0101 | First Trust International Equity Opport... | International IPO Equity    | FLAT |
| 78 | EIS  | 0.0097 | iShares MSCI Israel ETF                    | Israel Equity               | LONG |
| 40 | ITA  | 0.0095 | iShares U.S. Aerospace & Defense ETF       | Industrials                 | FLAT |

## Cluster 2 – Developed ex-US Equity

**Members:** 79 **Var explained:** 0.840 **Signal:** LONG since 2026-07-30 (19 days, 2.1%)

### Signal logic:

- Latest buy\_cross: 2026-07-30 – filter **PASS** (252d at date = +20.0%)
- Latest sell\_cross: 2026-06-24
- Current 252d return: +19.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-30 > last sell at 2026-06-24)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| International Equity          | 21        | 0.2771     |
| Developed ex-US Equity        | 9         | 0.1219     |
| International Dividend Equity | 6         | 0.0770     |
| Global ex-US Equity           | 5         | 0.0676     |
| Europe Equity                 | 4         | 0.0536     |
| Alternatives                  | 4         | 0.0483     |
| Eurozone Equity               | 3         | 0.0386     |
| Asia-Pacific Equity           | 2         | 0.0250     |
| Real Estate                   | 2         | 0.0232     |
| Global Equity                 | 2         | 0.0227     |
| Currency                      | 2         | 0.0185     |
| Commodity                     | 1         | 0.0137     |
| Global ex-US Small Cap Equity | 1         | 0.0133     |
| Germany Equity                | 1         | 0.0128     |
| France Equity                 | 1         | 0.0128     |
| United Kingdom Equity         | 1         | 0.0126     |
| Italy Equity                  | 1         | 0.0126     |
| Europe Financials             | 1         | 0.0124     |
| Sweden Equity                 | 1         | 0.0124     |
| Australia Equity              | 1         | 0.0123     |
| Netherlands Equity            | 1         | 0.0123     |
| Spain Equity                  | 1         | 0.0120     |

|                        |   |        |
|------------------------|---|--------|
| Canada Equity          | 1 | 0.0118 |
| Switzerland Equity     | 1 | 0.0117 |
| Austria Equity         | 1 | 0.0116 |
| Belgium Equity         | 1 | 0.0115 |
| Singapore Equity       | 1 | 0.0108 |
| Poland Equity          | 1 | 0.0107 |
| Consumer Discretionary | 1 | 0.0097 |
| New Zealand Equity     | 1 | 0.0094 |

**Replication basket** – VEA (cor=0.99) basket lead at lag 1: -0.017

| ticker | rep_wt | name                                | category               | etf_signal |
|--------|--------|-------------------------------------|------------------------|------------|
| VEA    | 0.911  | Vanguard FTSE Developed Markets ETF | Developed ex-US Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 65 | IEFA   | 0.0138 | iShares Core MSCI EAFE ETF                 | Developed ex-US Equity      | LONG       |
| 3  | EFA    | 0.0137 | iShares MSCI EAFE ETF                      | Developed ex-US Equity      | LONG       |
| 4  | VEA    | 0.0137 | Vanguard FTSE Developed Markets ETF        | Developed ex-US Equity      | LONG       |
| 51 | GSIE   | 0.0137 | Goldman Sachs ActiveBeta International ... | Commodity                   | LONG       |
| 54 | ESGD   | 0.0137 | iShares ESG Aware MSCI EAFE ETF            | International Equity        | LONG       |
| 66 | SCHF   | 0.0137 | Schwab International Equity ETF            | Developed ex-US Equity      | LONG       |
| 67 | SPDW   | 0.0137 | State Street SPDR Portfolio Developed W... | Developed ex-US Equity      | LONG       |
| 23 | DWM    | 0.0136 | WisdomTree International Equity Fund       | International Dividend E... | LONG       |
| 36 | TLTD   | 0.0136 | FlexShares Morningstar Developed Market... | International Equity        | LONG       |
| 43 | INTF   | 0.0136 | iShares International Equity Factor ETF    | International Equity        | LONG       |
| 61 | IXUS   | 0.0136 | iShares Core MSCI Total International S... | Global ex-US Equity         | LONG       |
| 7  | IEV    | 0.0135 | iShares Europe ETF                         | Europe Equity               | LONG       |
| 30 | PXF    | 0.0135 | Invesco RAFI Developed Markets ex-U.S. ... | International Equity        | LONG       |
| 37 | JPIN   | 0.0135 | JPMorgan Diversified Return Internation... | International Equity        | LONG       |
| 40 | ACWX   | 0.0135 | iShares MSCI ACWI ex U.S. ETF              | Global ex-US Equity         | LONG       |
| 41 | CWI    | 0.0135 | State Street SPDR MSCI ACWI ex-US ETF      | Global ex-US Equity         | LONG       |
| 44 | IQLT   | 0.0135 | iShares MSCI Intl Quality Factor ETF       | International Equity        | LONG       |
| 46 | QEFA   | 0.0135 | State Street SPDR MSCI EAFE StrategicFa... | International Equity        | LONG       |
| 50 | FNDF   | 0.0135 | Schwab Fundamental International Equity... | International Equity        | LONG       |
| 60 | IEUR   | 0.0135 | iShares Core MSCI Europe ETF               | Europe Equity               | LONG       |
| 62 | VEU    | 0.0135 | Vanguard FTSE All World Ex US ETF          | Global ex-US Equity         | LONG       |
| 63 | VGK    | 0.0135 | Vanguard FTSEEuropean ETF                  | Europe Equity               | LONG       |
| 64 | VXUS   | 0.0135 | Vanguard Total International Stock ETF     | Global ex-US Equity         | LONG       |
| 33 | SCZ    | 0.0134 | iShares MSCI EAFE Small-Cap ETF            | Developed ex-US Equity      | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 34 | EFV  | 0.0134 | iShares MSCI EAFE Value ETF                | Developed ex-US Equity      | LONG |
| 35 | FNDC | 0.0134 | Schwab Fundamental International Small ... | International Equity        | LONG |
| 48 | RODM | 0.0134 | Hartford Multifactor Developed Markets ... | International Equity        | LONG |
| 49 | VYMI | 0.0134 | Vanguard International High Dividend Yi... | International Dividend E... | LONG |
| 58 | SCHC | 0.0134 | Schwab International Small-Cap Equity E... | Developed ex-US Equity      | LONG |
| 21 | EZU  | 0.0133 | iShares MSCI Eurozone ETF                  | Eurozone Equity             | LONG |
| 26 | IQDF | 0.0133 | FlexShares International Quality Divide... | International Equity        | LONG |
| 59 | VSS  | 0.0133 | Vanguard FTSE All-Wld ex-US SmCp Idx ETF   | Global ex-US Small Cap E... | LONG |
| 9  | DLS  | 0.0132 | WisdomTree International SmallCap Fund     | International Equity        | LONG |
| 31 | EFG  | 0.0132 | iShares MSCI EAFE Growth ETF               | Developed ex-US Equity      | LONG |
| 45 | IVLU | 0.0132 | iShares MSCI Intl Value Factor ETF         | International Equity        | LONG |
| 71 | HFXI | 0.0132 | NYLI FTSE International Equity Currency... | International Equity        | LONG |
| 5  | FEP  | 0.0131 | First Trust Europe AlphaDEX Fund           | Europe Equity               | LONG |
| 47 | FEZ  | 0.0131 | State Street SPDR EURO STOXX 50 ETF        | Eurozone Equity             | LONG |
| 56 | VIGI | 0.0131 | Vanguard International Dividend Appreci... | International Dividend E... | LONG |
| 42 | IMTM | 0.0130 | iShares MSCI Intl Momentum Factor ETF      | International Equity        | LONG |
| 32 | GWX  | 0.0129 | State Street SPDR S&P International Sma... | International Equity        | LONG |
| 2  | EWG  | 0.0128 | iShares MSCI Germany Index Fund            | Germany Equity              | LONG |
| 6  | VIDI | 0.0128 | Vident International Equity Strategy ETF   | International Equity        | LONG |
| 17 | EWQ  | 0.0128 | iShares MSCI France Index Fund             | France Equity               | FLAT |
| 76 | IPKW | 0.0128 | Invesco International BuyBack Achievers... | International Equity        | LONG |
| 22 | IDV  | 0.0127 | iShares International Select Dividend E... | International Dividend E... | LONG |
| 19 | EWU  | 0.0126 | iShares MSCI United Kingdom ETF            | United Kingdom Equity       | LONG |
| 55 | HDEF | 0.0126 | Xtrackers MSCI EAFE High Dividend Yield... | International Equity        | LONG |
| 79 | EWI  | 0.0126 | iShares MSCI Italy ETF                     | Italy Equity                | FLAT |
| 24 | EPP  | 0.0125 | iShares MSCI Pacific Ex-Japan Index Fund   | Asia-Pacific Equity         | LONG |
| 52 | IDLV | 0.0125 | Invesco S&P International Developed Low... | International Equity        | LONG |
| 78 | BBAX | 0.0125 | JPMorgan BetaBuilders Developed Asia Pa... | Asia-Pacific Equity         | LONG |
| 25 | EUFN | 0.0124 | iShares MSCI Europe Financials ETF         | Europe Financials           | LONG |
| 70 | EWD  | 0.0124 | iShares MSCI Sweden ETF                    | Sweden Equity               | LONG |
| 14 | EWA  | 0.0123 | iShares MSCI Australia Index Fund          | Australia Equity            | LONG |
| 38 | DBEF | 0.0123 | Xtrackers MSCI EAFE Hedged Equity ETF      | Alternatives                | FLAT |
| 39 | HEFA | 0.0123 | iShares Currency Hedged MSCI EAFE ETF      | Alternatives                | FLAT |
| 53 | EFAV | 0.0123 | iShares MSCI EAFE Min Vol Factor ETF       | International Equity        | LONG |
| 74 | EWN  | 0.0123 | iShares MSCI Netherlands Index Fund        | Netherlands Equity          | LONG |
| 13 | HEZU | 0.0122 | iShares Currency Hedged MSCI Eurozone E... | Eurozone Equity             | LONG |
| 29 | DWX  | 0.0122 | State Street SPDR S&P International Div... | International Dividend E... | LONG |
| 16 | EWP  | 0.0120 | iShares MSCI Spain ETF                     | Spain Equity                | FLAT |
| 11 | IHDG | 0.0119 | WisdomTree International Hedged Quality... | Alternatives                | FLAT |
| 12 | PID  | 0.0119 | Invesco International Dividend Achiever... | International Dividend E... | FLAT |
| 10 | HEDJ | 0.0118 | WisdomTree Europe Hedged Equity Fund       | Alternatives                | FLAT |
| 20 | EWC  | 0.0118 | iShares MSCI Canada Index Fund             | Canada Equity               | LONG |
| 57 | VNQI | 0.0118 | Vanguard Global ex-U.S. Real Estate ETF    | Real Estate                 | LONG |
| 15 | EWL  | 0.0117 | iShares MSCI Switzerland ETF               | Switzerland Equity          | LONG |
| 27 | NFRA | 0.0117 | FlexShares STOXX Global Broad Infrastru... | Global Equity               | LONG |

|    |      |        |                                            |                        |      |
|----|------|--------|--------------------------------------------|------------------------|------|
| 73 | EWO  | 0.0116 | iShares MSCI Austria ETF                   | Austria Equity         | LONG |
| 77 | EWK  | 0.0115 | iShares MSCI Belgium ETF                   | Belgium Equity         | LONG |
| 8  | RWX  | 0.0114 | State Street SPDR Dow Jones Internation... | Real Estate            | FLAT |
| 28 | IGF  | 0.0111 | iShares Global Infrastructure ETF          | Global Equity          | FLAT |
| 18 | EWS  | 0.0108 | iShares MSCI Singapore ETF                 | Singapore Equity       | LONG |
| 69 | EPOL | 0.0107 | iShares MSCI Poland ETF                    | Poland Equity          | LONG |
| 68 | FXA  | 0.0100 | Invesco CurrencyShares Australian Dolla... | Currency               | LONG |
| 1  | ESPO | 0.0097 | VanEck Video Gaming and eSports ETF        | Consumer Discretionary | FLAT |
| 72 | ENZL | 0.0094 | iShares MSCI New Zealand ETF               | New Zealand Equity     | LONG |
| 75 | FXC  | 0.0085 | Invesco CurrencyShares Canadian Dollar ... | Currency               | LONG |

### Cluster 3 – U.S. Mid Cap Equity

**Members:** 72   **Var explained:** 0.897   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-10 – filter **PASS** (252d at date = +40.0%)
- Latest sell\_cross: 2026-08-25
- Current 252d return: +20.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-25 more recent than last effective buy 2026-04-10)



## Category breakdown

| category                              | n_members | abs_weight |
|---------------------------------------|-----------|------------|
| U.S. Small Cap Equity                 | 10        | 0.1440     |
| U.S. Mid Cap Equity                   | 6         | 0.0863     |
| U.S. Small Cap Value                  | 6         | 0.0859     |
| U.S. Small Cap Growth                 | 6         | 0.0841     |
| Consumer Discretionary                | 4         | 0.0517     |
| Industrials                           | 4         | 0.0498     |
| U.S. Mid Cap Value                    | 3         | 0.0432     |
| U.S. Mid Cap Growth                   | 3         | 0.0420     |
| Leveraged / Inverse                   | 2         | 0.0276     |
| Dividend Equity                       | 2         | 0.0273     |
| U.S. Small Cap Low Volatility         | 2         | 0.0261     |
| U.S. Small Cap Fundamental Equity     | 1         | 0.0146     |
| U.S. Small Cap ESG Equity             | 1         | 0.0145     |
| Unknown                               | 1         | 0.0145     |
| U.S. Mid/Small Cap Equity             | 1         | 0.0145     |
| U.S. Small/Mid Cap Fundamental Equity | 1         | 0.0145     |
| Mid Cap Value Equities                | 1         | 0.0145     |
| U.S. Small Cap Multifactor Equity     | 1         | 0.0144     |
| Small Cap Value Equities              | 1         | 0.0142     |
| U.S. Small/Mid Cap Equity             | 1         | 0.0142     |
| U.S. Mid Cap Dividend Equity          | 1         | 0.0141     |
| U.S. ESG Equity                       | 1         | 0.0141     |
| U.S. Extended Market Equity           | 1         | 0.0141     |
| U.S. Small Cap Dividend Equity        | 1         | 0.0139     |
| Value Equity                          | 1         | 0.0139     |
| U.S. Equity                           | 1         | 0.0139     |
| U.S. Mid Cap Quality Equity           | 1         | 0.0139     |
| Growth Equity                         | 1         | 0.0139     |
| U.S. Small Cap Industrials            | 1         | 0.0136     |
| U.S. Small Cap Momentum Equity        | 1         | 0.0136     |
| U.S. Micro Cap Equity                 | 1         | 0.0134     |
| U.S. Small Cap Consumer Discretionary | 1         | 0.0131     |
| U.S. Dividend Equity                  | 1         | 0.0131     |
| U.S. Mid Cap Low Volatility           | 1         | 0.0124     |
| Real Estate                           | 1         | 0.0113     |

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.013

| ticker | rep_wt | name | category | etf_signal |
|--------|--------|------|----------|------------|
|--------|--------|------|----------|------------|

IJH      1.015      iShares Core S&P Mid-Cap ETF      U.S. Mid Cap Equity      FLAT

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 47 | FNDA   | 0.0146 | Schwab Fundamental U.S. Small Company E... | U.S. Small Cap Fundament... | FLAT       |
| 8  | FNX    | 0.0145 | First Trust Mid Cap Core AlphaDEX Fund     | U.S. Mid/Small Cap Equity   | FLAT       |
| 11 | NUSC   | 0.0145 | Nuveen ESG Small-Cap ETF                   | U.S. Small Cap ESG Equity   | FLAT       |
| 12 | PRFZ   | 0.0145 | Invesco RAFI US 1500 Small-Mid ETF         | U.S. Small/Mid Cap Funda... | FLAT       |
| 15 | MDY    | 0.0145 | State Street SPDR S&P MIDCAP 400 ETF Tr... | U.S. Mid Cap Equity         | FLAT       |
| 23 | IVOO   | 0.0145 | Vanguard S&P Mid-Cap 400 ETF               |                             | FLAT       |
| 24 | VIOO   | 0.0145 | Vanguard S&P Small-Cap 600 ETF             | U.S. Small Cap Equity       | FLAT       |
| 27 | IJR    | 0.0145 | iShares Core S&P Small-Cap ETF             | U.S. Small Cap Equity       | FLAT       |
| 30 | IJH    | 0.0145 | iShares Core S&P Mid-Cap ETF               | U.S. Mid Cap Equity         | FLAT       |
| 31 | SPMD   | 0.0145 | State Street SPDR Portfolio S&P 400 Mid... | U.S. Mid Cap Equity         | FLAT       |
| 32 | SPSM   | 0.0145 | State Street SPDR Portfolio S&P 600 Sma... | U.S. Small Cap Equity       | FLAT       |
| 33 | VB     | 0.0145 | Vanguard Small-Cap ETF                     | U.S. Small Cap Equity       | FLAT       |
| 44 | EZM    | 0.0145 | WisdomTree U.S. MidCap Fund                | Mid Cap Value Equities      | FLAT       |
| 59 | VBR    | 0.0145 | Vanguard Small-Cap Value ETF               | U.S. Small Cap Value        | FLAT       |
| 4  | IWM    | 0.0144 | iShares Russell 2000 Index Fund            | U.S. Small Cap Equity       | LONG       |
| 9  | FYX    | 0.0144 | First Trust Small Cap Core AlphaDEX Fund   | U.S. Small Cap Equity       | FLAT       |
| 13 | SMLF   | 0.0144 | iShares U.S. Small-Cap Equity Factor ETF   | U.S. Small Cap Multifact... | LONG       |
| 16 | IJT    | 0.0144 | iShares S&P SmallCap 600 Growth ETF        | U.S. Small Cap Growth       | FLAT       |
| 18 | VIOG   | 0.0144 | Vanguard S&P Small-Cap 600 Growth ETF      | U.S. Small Cap Growth       | FLAT       |
| 21 | SLYG   | 0.0144 | State Street SPDR S&P 600 Small Cap Gro... | U.S. Small Cap Growth       | FLAT       |
| 25 | VTWO   | 0.0144 | Vanguard Russell 2000 ETF                  | U.S. Small Cap Equity       | LONG       |
| 34 | SCHA   | 0.0144 | Schwab U.S. Small-Cap ETF                  | U.S. Small Cap Equity       | LONG       |
| 50 | IWN    | 0.0144 | iShares Russell 2000 Value ETF             | U.S. Small Cap Value        | FLAT       |
| 52 | IJJ    | 0.0144 | iShares S&P Mid-Cap 400 Value ETF          | U.S. Mid Cap Value          | FLAT       |
| 54 | IVOV   | 0.0144 | Vanguard S&P Mid-Cap 400 Value ETF         | U.S. Mid Cap Value          | FLAT       |
| 55 | MDYV   | 0.0144 | State Street SPDR S&P 400 Mid Cap Value... | U.S. Mid Cap Value          | FLAT       |
| 5  | JHMM   | 0.0143 | John Hancock Multifactor Mid Cap ETF       | U.S. Mid Cap Equity         | FLAT       |
| 7  | SCHM   | 0.0143 | Schwab U.S. Mid Cap ETF                    | U.S. Mid Cap Equity         | LONG       |
| 53 | IJS    | 0.0143 | iShares S&P SmallCap 600 Value ETF         | U.S. Small Cap Value        | LONG       |
| 56 | SLYV   | 0.0143 | State Street SPDR S&P 600 Small Cap Val... | U.S. Small Cap Value        | FLAT       |
| 58 | VTWV   | 0.0143 | Vanguard Russell 2000 Value ETF            | U.S. Small Cap Value        | FLAT       |
| 65 | RWK    | 0.0143 | Invesco S&P MidCap 400 Revenue ETF         | U.S. Mid Cap Equity         | FLAT       |
| 43 | EES    | 0.0142 | WisdomTree U.S. SmallCap Fund              | Small Cap Value Equities    | FLAT       |
| 45 | PSC    | 0.0142 | Principal U.S. Small-Cap ETF               | U.S. Small Cap Equity       | FLAT       |
| 57 | VIOV   | 0.0142 | Vanguard S&P Small-Cap 600 Value ETF       | U.S. Small Cap Value        | FLAT       |
| 61 | RWJ    | 0.0142 | Invesco S&P SmallCap 600 Revenue ETF       | U.S. Small Cap Equity       | FLAT       |
| 71 | SDVY   | 0.0142 | First Trust SMID Cap Rising Dividend Ac... | Dividend Equity             | FLAT       |
| 72 | ISMD   | 0.0142 | Inspire Small/Mid Cap ETF                  | U.S. Small/Mid Cap Equity   | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 29 | VXF  | 0.0141 | Vanguard Extended Market ETF               | U.S. Extended Market Equ... | LONG |
| 42 | DON  | 0.0141 | WisdomTree U.S. MidCap Dividend Fund       | U.S. Mid Cap Dividend Eq... | FLAT |
| 60 | ETHO | 0.0141 | Amplify Etho Climate Leadership U.S. ETF   | U.S. ESG Equity             | FLAT |
| 17 | IJK  | 0.0140 | iShares S&P Mid-Cap 400 Growth ETF         | U.S. Mid Cap Growth         | FLAT |
| 19 | IVOG | 0.0140 | Vanguard S&P Mid-Cap 400 Growth ETF        | U.S. Mid Cap Growth         | FLAT |
| 20 | MDYG | 0.0140 | State Street SPDR S&P 400 Mid Cap Growt... | U.S. Mid Cap Growth         | FLAT |
| 28 | ISCG | 0.0139 | iShares Morningstar Small-Cap Growth ETF   | Growth Equity               | LONG |
| 41 | DES  | 0.0139 | WisdomTree U.S. SmallCap Dividend Fund     | U.S. Small Cap Dividend ... | FLAT |
| 63 | XMHQ | 0.0139 | Invesco S&P MidCap Quality ETF             | U.S. Mid Cap Quality Equ... | FLAT |
| 64 | RFV  | 0.0139 | Invesco S&P MidCap 400 Pure Value ETF      | Leveraged / Inverse         | FLAT |
| 66 | OMFS | 0.0139 | Invesco Russell 2000 Dynamic Multifact...  | U.S. Equity                 | FLAT |
| 70 | FYT  | 0.0139 | First Trust Small Cap Value AlphaDEX Fu... | Value Equity                | FLAT |
| 14 | IWO  | 0.0137 | iShares Russell 2000 Growth Fund           | U.S. Small Cap Growth       | LONG |
| 22 | VTWG | 0.0137 | Vanguard Russell 2000 Growth ETF           | U.S. Small Cap Growth       | LONG |
| 67 | RZV  | 0.0137 | Invesco S&P SmallCap 600 Pure Value ETF    | Leveraged / Inverse         | FLAT |
| 26 | VBK  | 0.0136 | Vanguard Small-Cap Growth ETF              | U.S. Small Cap Growth       | LONG |
| 62 | PSCI | 0.0136 | Invesco S&P SmallCap Industrials ETF       | U.S. Small Cap Industria... | FLAT |
| 69 | XSMO | 0.0136 | Invesco S&P SmallCap Momentum ETF          | U.S. Small Cap Momentum ... | FLAT |
| 35 | FXD  | 0.0135 | First Trust Cons. Discret. AlphaDEX        | Consumer Discretionary      | FLAT |
| 10 | IWC  | 0.0134 | iShares Microcap ETF                       | U.S. Micro Cap Equity       | LONG |
| 39 | RSPD | 0.0132 | Invesco S&P 500 Equal Weight Consumer D... | Consumer Discretionary      | FLAT |
| 51 | SMMV | 0.0132 | iShares MSCI USA Small-Cap Min Vol Fact... | U.S. Small Cap Low Volat... | LONG |
| 6  | REGL | 0.0131 | ProShares S&P MidCap 400 Dividend Arist... | Dividend Equity             | FLAT |
| 37 | FTXR | 0.0131 | First Trust Nasdaq Transportation ETF      | Industrials                 | FLAT |
| 40 | SMDV | 0.0131 | ProShares Russell 2000 Dividend Growers... | U.S. Dividend Equity        | FLAT |
| 68 | PSCD | 0.0131 | Invesco S&P SmallCap Consumer Discretio... | U.S. Small Cap Consumer ... | FLAT |
| 46 | XTN  | 0.0129 | State Street SPDR S&P Transportation ETF   | Industrials                 | FLAT |
| 49 | XSLV | 0.0128 | Invesco S&P SmallCap Low Volatility ETF    | U.S. Small Cap Low Volat... | FLAT |
| 1  | XRT  | 0.0126 | State Street SPDR S&P Retail ETF           | Consumer Discretionary      | FLAT |
| 38 | IYT  | 0.0124 | iShares U.S. Transportation ETF            | Industrials                 | LONG |
| 48 | XMLV | 0.0124 | Invesco S&P MidCap Low Volatility ETF      | U.S. Mid Cap Low Volatil... | FLAT |
| 36 | PEJ  | 0.0123 | Invesco Leisure and Entertainment ETF      | Consumer Discretionary      | FLAT |
| 2  | JETS | 0.0114 | U.S. Global Jets ETF                       | Industrials                 | FLAT |
| 3  | REM  | 0.0113 | iShares Mortgage Real Estate ETF           | Real Estate                 | LONG |

#### Cluster 4 – U.S. Large Cap Value

**Members:** 60 **Var explained:** 0.890 **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +26.6%)

- Latest sell\_cross: 2026-07-23
- Current 252d return: +19.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-13)



### Category breakdown

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| U.S. Large Cap Value         | 10        | 0.1728     |
| U.S. Dividend Equity         | 8         | 0.1360     |
| Industrials                  | 7         | 0.1142     |
| Materials                    | 4         | 0.0627     |
| U.S. Mid Cap Equity          | 3         | 0.0516     |
| U.S. Multifactor Equity      | 3         | 0.0512     |
| U.S. Value Factor Equity     | 3         | 0.0502     |
| Utilities                    | 3         | 0.0468     |
| U.S. Fundamental Equity      | 2         | 0.0348     |
| U.S. Equal Weight Equity     | 2         | 0.0347     |
| U.S. Mid Cap Value           | 2         | 0.0346     |
| U.S. Equity                  | 2         | 0.0343     |
| Consumer Discretionary       | 2         | 0.0272     |
| Large Cap Blend Equities     | 1         | 0.0173     |
| U.S. Size Factor Equity      | 1         | 0.0172     |
| U.S. Revenue Weighted Equity | 1         | 0.0171     |
| Value Equity                 | 1         | 0.0170     |
| U.S. Mega Cap Value          | 1         | 0.0169     |

|                        |   |        |
|------------------------|---|--------|
| U.S. Buyback Equity    | 1 | 0.0168 |
| U.S. Large Cap Equity  | 1 | 0.0167 |
| U.S. Moat Equity       | 1 | 0.0162 |
| Communication Services | 1 | 0.0137 |

**Replication basket** – VTV (cor=0.97) basket lead at lag 1: -0.003

| ticker | rep_wt | name               | category             | etf_signal |
|--------|--------|--------------------|----------------------|------------|
| VTV    | 1.092  | Vanguard Value ETF | U.S. Large Cap Value | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 3  | RSP    | 0.0175 | Invesco S&P 500 Equal Weight ETF           | U.S. Equal Weight Equity    | LONG       |
| 18 | CFA    | 0.0175 | VictoryShares US 500 Volatility Wtd ETF    | U.S. Multifactor Equity     | FLAT       |
| 26 | IWD    | 0.0175 | iShares Russell 1000 Value ETF             | U.S. Large Cap Value        | LONG       |
| 36 | VONV   | 0.0175 | Vanguard Russell 1000 Value ETF            | U.S. Large Cap Value        | LONG       |
| 15 | PRF    | 0.0174 | Invesco RAFI US 1000 ETF                   | U.S. Fundamental Equity     | LONG       |
| 22 | FNDX   | 0.0174 | Schwab Fundamental U.S. Large Company E... | U.S. Fundamental Equity     | LONG       |
| 29 | JPUS   | 0.0174 | JPMorgan Diversified Return U.S. Equity... | U.S. Multifactor Equity     | FLAT       |
| 42 | SCHV   | 0.0174 | Schwab U.S. Large-Cap Value ETF            | U.S. Large Cap Value        | LONG       |
| 48 | IWS    | 0.0174 | iShares Russell Mid-Cap Value ETF          | U.S. Mid Cap Value          | FLAT       |
| 6  | FEX    | 0.0173 | First Trust Large Cap Core AlphaDEX Fund   | Large Cap Blend Equities    | LONG       |
| 39 | ILCV   | 0.0173 | iShares Morningstar Large-Cap Value ETF    | U.S. Large Cap Value        | FLAT       |
| 41 | IUSV   | 0.0173 | iShares Core S&P U.S. Value ETF            | U.S. Large Cap Value        | FLAT       |
| 59 | JPME   | 0.0173 | JPMorgan Diversified Return U.S. Mid Ca... | U.S. Equity                 | FLAT       |
| 23 | EQAL   | 0.0172 | Invesco Russell 1000 Equal Weight ETF      | U.S. Equal Weight Equity    | FLAT       |
| 27 | IWR    | 0.0172 | iShares Russell Mid-Cap ETF                | U.S. Mid Cap Equity         | FLAT       |
| 28 | IVE    | 0.0172 | iShares S&P 500 Value ETF                  | U.S. Large Cap Value        | FLAT       |
| 30 | SIZE   | 0.0172 | iShares MSCI USA Size Factor ETF           | U.S. Size Factor Equity     | FLAT       |
| 34 | VOOV   | 0.0172 | Vanguard S&P 500 Value ETF                 | U.S. Large Cap Value        | FLAT       |
| 40 | IMCB   | 0.0172 | iShares Morningstar Mid-Cap ETF            | U.S. Mid Cap Equity         | FLAT       |
| 43 | SPYV   | 0.0172 | State Street SPDR Portfolio S&P 500 Val... | U.S. Large Cap Value        | FLAT       |
| 45 | VTV    | 0.0172 | Vanguard Value ETF                         | U.S. Large Cap Value        | LONG       |
| 49 | VOE    | 0.0172 | Vanguard Mid-Cap Value ETF                 | U.S. Mid Cap Value          | FLAT       |
| 52 | DTD    | 0.0172 | WisdomTree U.S. Total Dividend Fund        | U.S. Dividend Equity        | LONG       |
| 16 | RWL    | 0.0171 | Invesco S&P 500 Revenue ETF                | U.S. Revenue Weighted Eq... | LONG       |
| 20 | NULV   | 0.0171 | Nuveen ESG Large-Cap Value ETF             | U.S. Large Cap Value        | FLAT       |
| 44 | VO     | 0.0171 | Vanguard Mid-Cap ETF                       | U.S. Mid Cap Equity         | FLAT       |
| 54 | DGRO   | 0.0171 | iShares Core Dividend Growth ETF           | U.S. Dividend Equity        | LONG       |
| 60 | JVAL   | 0.0171 | JPMorgan U.S. Value Factor ETF             | U.S. Value Factor Equity    | LONG       |

|    |      |        |                                            |                          |      |
|----|------|--------|--------------------------------------------|--------------------------|------|
| 12 | IYJ  | 0.0170 | iShares U.S. Industrials ETF               | Industrials              | FLAT |
| 17 | AIVL | 0.0170 | WisdomTree U.S. AI Enhanced Value Fund     | U.S. Value Factor Equity | LONG |
| 24 | IWX  | 0.0170 | iShares Russell Top 200 Value ETF          | Value Equity             | LONG |
| 25 | ONEV | 0.0170 | State Street SPDR Russell 1000 Low Vola... | U.S. Equity              | FLAT |
| 38 | VYM  | 0.0170 | Vanguard High Dividend Yield ETF           | U.S. Dividend Equity     | FLAT |
| 50 | PFM  | 0.0170 | Invesco Dividend Achievers ETF             | U.S. Dividend Equity     | FLAT |
| 51 | DLN  | 0.0170 | WisdomTree U.S. LargeCap Dividend Fund     | U.S. Dividend Equity     | LONG |
| 55 | VIG  | 0.0170 | Vanguard Div Appreciation ETF              | U.S. Dividend Equity     | LONG |
| 37 | MGV  | 0.0169 | Vanguard Mega Cap Value ETF                | U.S. Mega Cap Value      | FLAT |
| 5  | PKW  | 0.0168 | Invesco BuyBack Achievers ETF              | U.S. Buyback Equity      | LONG |
| 10 | RDVY | 0.0168 | First Trust Rising Dividend Achievers E... | U.S. Dividend Equity     | FLAT |
| 21 | FDVV | 0.0168 | Fidelity High Dividend ETF                 | U.S. Dividend Equity     | FLAT |
| 4  | FXR  | 0.0167 | First Trust Industrials AlphaDEX           | Industrials              | FLAT |
| 53 | DIA  | 0.0167 | State Street SPDR Dow Jones Industrial ... | U.S. Large Cap Equity    | FLAT |
| 33 | VIS  | 0.0166 | Vanguard Industrials ETF                   | Industrials              | FLAT |
| 35 | FIDU | 0.0166 | Fidelity MSCI Industrials Index ETF        | Industrials              | FLAT |
| 1  | XLI  | 0.0164 | State Street Industrial Select Sector S... | Industrials              | FLAT |
| 19 | CFO  | 0.0164 | VictoryShares US 500 Enhanced Volatilit... | U.S. Multifactor Equity  | FLAT |
| 11 | MOAT | 0.0162 | VanEck Morningstar Wide Moat ETF           | U.S. Moat Equity         | FLAT |
| 56 | PAVE | 0.0162 | Global X U.S. Infrastructure Developmen... | Industrials              | FLAT |
| 31 | VLUE | 0.0161 | iShares MSCI USA Value Factor ETF          | U.S. Value Factor Equity | LONG |
| 9  | FIW  | 0.0160 | First Trust Water ETF                      | Utilities                | FLAT |
| 32 | VAW  | 0.0160 | Vanguard Materials ETF                     | Materials                | LONG |
| 2  | XLB  | 0.0158 | State Street Materials Select Sector SP... | Materials                | LONG |
| 7  | PHO  | 0.0158 | Invesco Water Resources ETF                | Utilities                | FLAT |
| 14 | RSPM | 0.0156 | Invesco S&P 500 Equal Weight Materials ... | Materials                | LONG |
| 13 | IYM  | 0.0152 | iShares U.S. Basic Materials ETF           | Materials                | LONG |
| 8  | CGW  | 0.0150 | Invesco S&P Global Water Index ETF         | Utilities                | LONG |
| 58 | PKB  | 0.0148 | Invesco Building & Construction ETF        | Industrials              | FLAT |
| 47 | XHB  | 0.0142 | State Street SPDR S&P Homebuilders ETF     | Consumer Discretionary   | FLAT |
| 57 | IYZ  | 0.0137 | iShares U.S. Telecommunications ETF        | Communication Services   | LONG |
| 46 | ITB  | 0.0130 | iShares U.S. Home Construction ETF         | Consumer Discretionary   | LONG |

## Cluster 5 – Aggregate Fixed Income

**Members:** 58 **Var explained:** 0.831 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell\_cross: 2026-07-07
- Current 252d return: +1.2% – filter **ALLOWS** new entries today

- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Fixed Income                    | 19        | 0.3285     |
| Investment Grade Corporate Bond | 13        | 0.2303     |
| Aggregate Fixed Income          | 6         | 0.1120     |
| Mortgage-Backed Fixed Income    | 6         | 0.0982     |
| Active Fixed Income             | 5         | 0.0891     |
| Cash / Ultra-Short Fixed Income | 3         | 0.0470     |
| International Fixed Income      | 2         | 0.0313     |
| Target Maturity Bond            | 2         | 0.0305     |
| Agency Fixed Income             | 1         | 0.0166     |
| Taxable Municipal Bond          | 1         | 0.0165     |

**Replication basket** – BND (cor=0.99) basket lead at lag 1: 0.006

| ticker | rep_wt | name                           | category               | etf_signal |
|--------|--------|--------------------------------|------------------------|------------|
| BND    | 1.008  | Vanguard Total Bond Market ETF | Aggregate Fixed Income | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                        | category                    | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------------------|------------|
| 21 | BIV    | 0.0188 | Vanguard Intermediate-Term Bond ETF         | Fixed Income                | FLAT       |
| 26 | AGG    | 0.0188 | iShares Core U.S. Aggregate Bond ETF        | Aggregate Fixed Income      | FLAT       |
| 28 | BND    | 0.0188 | Vanguard Total Bond Market ETF              | Aggregate Fixed Income      | FLAT       |
| 30 | SCHZ   | 0.0188 | Schwab US Aggregate Bond ETF                | Aggregate Fixed Income      | FLAT       |
| 20 | IUSB   | 0.0187 | iShares Core Universal USD Bond ETF         | Aggregate Fixed Income      | FLAT       |
| 31 | SPAB   | 0.0187 | State Street SPDR Portfolio Aggregate B...  | Aggregate Fixed Income      | FLAT       |
| 2  | IEF    | 0.0185 | iShares 7-10 Year Treasury Bond ETF         | Fixed Income                | FLAT       |
| 17 | IGIB   | 0.0184 | iShares 5-10 Year Investment Grade Corp...  | Investment Grade Corpora... | FLAT       |
| 24 | VCIT   | 0.0184 | Vanguard Intermediate-Term Corporate Bo...  | Investment Grade Corpora... | FLAT       |
| 7  | FBND   | 0.0183 | Fidelity Total Bond ETF                     | Active Fixed Income         | FLAT       |
| 43 | SPTI   | 0.0183 | State Street SPDR Portfolio Intermediat...  | Fixed Income                | FLAT       |
| 46 | SCHR   | 0.0183 | Schwab Intermediate-Term U.S. Treasury ...  | Fixed Income                | FLAT       |
| 47 | VGIT   | 0.0183 | Vanguard Intermediate-Term Treasury ETF     | Fixed Income                | FLAT       |
| 13 | AGGY   | 0.0182 | WisdomTree Yield Enhanced U.S. Aggregat...  | Aggregate Fixed Income      | FLAT       |
| 15 | SPIB   | 0.0182 | State Street SPDR Portfolio Intermediat...  | Investment Grade Corpora... | FLAT       |
| 6  | GTO    | 0.0181 | Invesco Total Return Bond ETF               | Active Fixed Income         | FLAT       |
| 11 | GVI    | 0.0181 | iShares Intermediate Government/Credit ...  | Fixed Income                | FLAT       |
| 27 | USIG   | 0.0181 | iShares Broad USD Investment Grade Corp...  | Investment Grade Corpora... | FLAT       |
| 40 | IEI    | 0.0181 | iShares 3-7 Year Treasury Bond ETF          | Fixed Income                | FLAT       |
| 50 | MBB    | 0.0181 | iShares MBS ETF                             | Mortgage-Backed Fixed In... | FLAT       |
| 12 | QLTA   | 0.0180 | iShares Aaa A Rated Corporate Bond ETF      | Investment Grade Corpora... | FLAT       |
| 51 | VMBS   | 0.0180 | Vanguard Mortgage-Backed Securities ETF     | Mortgage-Backed Fixed In... | FLAT       |
| 52 | SPMB   | 0.0180 | State Street SPDR Portfolio Mortgage Ba...  | Mortgage-Backed Fixed In... | FLAT       |
| 3  | LQD    | 0.0179 | iShares iBoxx \$ Investment Grade Corpor... | Investment Grade Corpora... | FLAT       |
| 5  | BOND   | 0.0179 | PIMCO Active Bond Exchange-Traded Fund ...  | Active Fixed Income         | FLAT       |
| 10 | CORP   | 0.0178 | PIMCO Investment Grade Corporate Bond I...  | Investment Grade Corpora... | FLAT       |
| 19 | ISTB   | 0.0178 | iShares Core 1-5 Year USD Bond ETF          | Fixed Income                | LONG       |
| 8  | FLCO   | 0.0177 | Franklin Investment Grade Corporate ETF     | Active Fixed Income         | FLAT       |
| 22 | BLV    | 0.0176 | Vanguard Long-Term Bond ETF                 | Fixed Income                | FLAT       |
| 41 | TLH    | 0.0176 | iShares 10-20 Year Treasury Bond ETF        | Fixed Income                | FLAT       |
| 45 | GOVT   | 0.0176 | iShares U.S. Treasury Bond ETF              | Fixed Income                | FLAT       |
| 35 | IGSB   | 0.0175 | iShares 1-5 Year Investment Grade Corpo...  | Investment Grade Corpora... | FLAT       |
| 37 | VCSH   | 0.0175 | Vanguard Short-Term Corporate Bond ETF      | Investment Grade Corpora... | FLAT       |
| 23 | BSV    | 0.0174 | Vanguard Short-Term Bond ETF                | Fixed Income                | LONG       |
| 49 | GNMA   | 0.0174 | iShares GNMA Bond ETF                       | Mortgage-Backed Fixed In... | FLAT       |
| 16 | SPLB   | 0.0172 | State Street SPDR Portfolio Long Term C...  | Investment Grade Corpora... | FLAT       |
| 18 | IGLB   | 0.0172 | iShares 10 Year Investment Grade Corpor...  | Investment Grade Corpora... | FLAT       |
| 44 | SPTL   | 0.0172 | State Street SPDR Portfolio Long Term T...  | Fixed Income                | FLAT       |
| 25 | VCLT   | 0.0171 | Vanguard Long-Term Corporate Bond ETF       | Investment Grade Corpora... | FLAT       |
| 36 | SLQD   | 0.0171 | iShares 0-5 Year Investment Grade Corpo...  | Investment Grade Corpora... | LONG       |
| 48 | VGLT   | 0.0171 | Vanguard Long-Term Treasury ETF             | Fixed Income                | FLAT       |
| 29 | TOTL   | 0.0170 | State Street DoubleLine Total Return Ta...  | Active Fixed Income         | FLAT       |
| 1  | TLT    | 0.0169 | iShares 20+ Year Treasury Bond ETF          | Fixed Income                | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 39 | AGZ  | 0.0166 | iShares Agency Bond ETF                    | Agency Fixed Income         | FLAT |
| 9  | BAB  | 0.0165 | Invesco Taxable Municipal Bond ETF         | Taxable Municipal Bond      | FLAT |
| 42 | EDV  | 0.0163 | Vanguard Extended Duration Treasury ETF    | Fixed Income                | FLAT |
| 34 | SPSB | 0.0162 | State Street SPDR Portfolio Short Term ... | Fixed Income                | LONG |
| 53 | SHY  | 0.0162 | iShares 1-3 Year Treasury Bond ETF         | Fixed Income                | LONG |
| 56 | VGSH | 0.0159 | Vanguard Short-Term Treasury ETF           | Cash / Ultra-Short Fixed... | LONG |
| 57 | BNDX | 0.0158 | Vanguard Total International Bond ETF      | International Fixed Inco... | FLAT |
| 55 | SCHO | 0.0156 | Schwab Short-Term U.S. Treasury ETF        | Cash / Ultra-Short Fixed... | LONG |
| 58 | IAGG | 0.0156 | iShares International Aggregate Bond Fu... | International Fixed Inco... | FLAT |
| 33 | BSCQ | 0.0155 | Invesco BulletShares 2026 Corporate Bon... | Target Maturity Bond        | LONG |
| 54 | SPTS | 0.0155 | State Street SPDR Portfolio Short Term ... | Cash / Ultra-Short Fixed... | LONG |
| 14 | IBDR | 0.0150 | iShares iBonds Dec 2026 Term Corporate ... | Target Maturity Bond        | LONG |
| 38 | CMBS | 0.0135 | iShares CMBS Bond ETF                      | Mortgage-Backed Fixed In... | LONG |
| 4  | LMBS | 0.0133 | First Trust Low Duration Opportunities ... | Mortgage-Backed Fixed In... | LONG |
| 32 | LDUR | 0.0121 | PIMCO Enhanced Low Duration Active Exch... | Fixed Income                | LONG |

### Cluster 6 – U.S. Large Cap Growth

**Members:** 38   **Var explained:** 0.893   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-08-10 – filter **PASS** (252d at date = +36.3%)
- Latest sell\_cross: 2026-08-26
- Current 252d return: +32.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-26 more recent than last effective buy 2026-08-10)



### Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| Technology                  | 19        | 0.4990     |
| U.S. Large Cap Growth       | 11        | 0.2988     |
| Growth Equity               | 2         | 0.0536     |
| U.S. Equity                 | 1         | 0.0273     |
| U.S. Mega Cap Growth        | 1         | 0.0270     |
| Options Income              | 1         | 0.0246     |
| U.S. Small Cap Technology   | 1         | 0.0243     |
| Trend-Following Equity      | 1         | 0.0242     |
| U.S. Trend-Following Equity | 1         | 0.0211     |

**Replication basket** – QQQ (cor=0.98) basket lead at lag 1: -0.030

| ticker | rep_wt | name                        | category              | etf_signal |
|--------|--------|-----------------------------|-----------------------|------------|
| QQQ    | 1.033  | Invesco QQQ Trust, Series 1 | U.S. Large Cap Growth | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 16 | IGM    | 0.0276 | iShares Expanded Tech Sector ETF           | Technology                  | FLAT       |
| 5  | QQQ    | 0.0275 | Invesco QQQ Trust, Series 1                | U.S. Large Cap Growth       | LONG       |
| 18 | IYW    | 0.0275 | iShares U.S. Technology ETF                | Technology                  | FLAT       |
| 24 | VGT    | 0.0275 | Vanguard Information Tech ETF              | Technology                  | FLAT       |
| 26 | FTEC   | 0.0275 | Fidelity MSCI Information Technology In... | Technology                  | FLAT       |
| 1  | XLK    | 0.0274 | State Street Technology Select Sector S... | Technology                  | FLAT       |
| 29 | ILCG   | 0.0274 | iShares Morningstar Large-Cap Growth ETF   | U.S. Large Cap Growth       | FLAT       |
| 17 | IXN    | 0.0273 | iShares Global Tech ETF                    | Technology                  | LONG       |
| 20 | ONEQ   | 0.0273 | Fidelity Nasdaq Composite Index ETF        | U.S. Equity                 | FLAT       |
| 22 | IWF    | 0.0272 | iShares Russell 1000 Growth Fund           | U.S. Large Cap Growth       | FLAT       |
| 23 | IVW    | 0.0272 | iShares S&P 500 Growth ETF                 | U.S. Large Cap Growth       | FLAT       |
| 25 | VOOG   | 0.0272 | Vanguard S&P 500 Growth ETF                | U.S. Large Cap Growth       | FLAT       |
| 27 | VONG   | 0.0272 | Vanguard Russell 1000 Growth ETF           | U.S. Large Cap Growth       | FLAT       |
| 30 | IUSG   | 0.0272 | iShares Core S&P U.S. Growth ETF           | U.S. Large Cap Growth       | FLAT       |
| 32 | SPYG   | 0.0272 | State Street SPDR Portfolio S&P 500 Gro... | U.S. Large Cap Growth       | FLAT       |
| 12 | XNTK   | 0.0271 | State Street SPDR NYSE Technology ETF      | Technology                  | LONG       |
| 33 | VUG    | 0.0271 | Vanguard Growth ETF                        | U.S. Large Cap Growth       | FLAT       |
| 19 | NULG   | 0.0270 | Nuveen ESG Large-Cap Growth ETF            | U.S. Large Cap Growth       | FLAT       |
| 21 | IWY    | 0.0270 | iShares Russell Top 200 Growth ETF         | Growth Equity               | FLAT       |
| 28 | MGK    | 0.0270 | Vanguard Mega Cap Growth ETF               | U.S. Mega Cap Growth        | FLAT       |
| 7  | QTEC   | 0.0269 | First Trust NASDAQ-100-Technology Secto... | Technology                  | LONG       |
| 31 | SCHG   | 0.0268 | Schwab U.S. Large-Cap Growth ETF           | U.S. Large Cap Growth       | LONG       |
| 4  | AIQ    | 0.0266 | Global X Artificial Intelligence & Tech... | Technology                  | LONG       |
| 11 | FXL    | 0.0266 | First Trust Technology AlphaDEX            | Technology                  | LONG       |
| 15 | PWB    | 0.0266 | Invesco Large Cap Growth ETF               | Growth Equity               | FLAT       |
| 3  | SMH    | 0.0259 | VanEck Semiconductor ETF                   | Technology                  | FLAT       |
| 9  | TDIV   | 0.0259 | First Trust NASDAQ Technology Dividend ... | Technology                  | FLAT       |
| 2  | SOXX   | 0.0256 | iShares PHLX SOX Semiconductor Sector I... | Technology                  | FLAT       |
| 38 | QTUM   | 0.0255 | Defiance Quantum ETF                       | Technology                  | FLAT       |
| 10 | XSD    | 0.0250 | State Street SPDR S&P Semiconductor ETF    | Technology                  | FLAT       |
| 36 | FTXL   | 0.0250 | First Trust Nasdaq Semiconductor ETF       | Technology                  | FLAT       |
| 8  | PSI    | 0.0249 | Invesco Semiconductors ETF                 | Technology                  | FLAT       |
| 6  | NXTG   | 0.0248 | First Trust Indxx NextG ETF                | Technology                  | LONG       |
| 14 | QYLD   | 0.0246 | Global X NASDAQ 100 Covered Call ETF       | Options Income              | LONG       |
| 35 | PSCT   | 0.0243 | Invesco S&P SmallCap Information Techno... | U.S. Small Cap Technology   | FLAT       |
| 37 | PTF    | 0.0243 | Invesco Dorsey Wright Technology Moment... | Technology                  | FLAT       |
| 13 | PTNQ   | 0.0242 | Pacer Trendpilot 100 ETF                   | Trend-Following Equity      | LONG       |
| 34 | PTLC   | 0.0211 | Pacer Trendpilot US Large Cap ETF          | U.S. Trend-Following Equ... | LONG       |

**Cluster 7 – Emerging Markets Equity**

**Members:** 30   **Var explained:** 0.798   **Signal:** LONG since 2026-08-12 (10 days, 1.3%)

**Signal logic:**

- Latest buy\_cross: 2026-08-12 – filter **PASS** (252d at date = +29.5%)
- Latest sell\_cross: 2026-07-07
- Current 252d return: +28.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-12 > last sell at 2026-07-07)

**Category breakdown**

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| Emerging Markets Equity | 19        | 0.6753     |
| Asia Equity             | 2         | 0.0703     |
| Asia ex-Japan Equity    | 1         | 0.0361     |
| Taiwan Equity           | 1         | 0.0313     |
| Global Dividend Equity  | 1         | 0.0299     |
| South Africa Equity     | 1         | 0.0297     |
| South Korea Equity      | 1         | 0.0294     |
| Latin America Equity    | 1         | 0.0275     |
| Mexico Equity           | 1         | 0.0245     |
| Brazil Equity           | 1         | 0.0233     |

|              |   |        |
|--------------|---|--------|
| Chile Equity | 1 | 0.0226 |
|--------------|---|--------|

**Replication basket** – IEMG (cor=0.99) basket lead at lag 1: -0.032

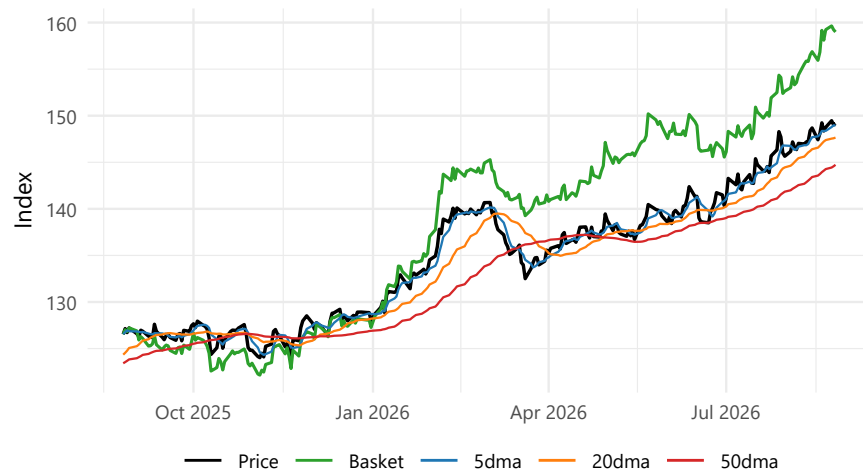
| ticker | rep_wt | name                                   | category                 | etf_signal |
|--------|--------|----------------------------------------|--------------------------|------------|
| IEMG   | 0.908  | iShares Core MSCI Emerging Markets ETF | Emerging Markets Equi... | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 22 | IEMG   | 0.0370 | iShares Core MSCI Emerging Markets ETF     | Emerging Markets Equity | LONG       |
| 3  | EEM    | 0.0369 | iShares MSCI Emerging Index Fund           | Emerging Markets Equity | LONG       |
| 24 | SPEM   | 0.0369 | State Street SPDR Portfolio Emerging Ma... | Emerging Markets Equity | LONG       |
| 4  | VWO    | 0.0368 | Vanguard FTSE Emerging Markets ETF         | Emerging Markets Equity | LONG       |
| 17 | GEM    | 0.0368 | Goldman Sachs ActiveBeta Emerging Marke... | Emerging Markets Equity | LONG       |
| 21 | ESGE   | 0.0368 | iShares ESG Aware MSCI EM ETF              | Emerging Markets Equity | LONG       |
| 23 | SCHE   | 0.0367 | Schwab Emerging Markets Equity ETF         | Emerging Markets Equity | LONG       |
| 19 | XSOE   | 0.0365 | WisdomTree Emerging Markets Ex-State Ow... | Emerging Markets Equity | LONG       |
| 16 | EMGF   | 0.0364 | iShares Emerging Markets Equity Factor ... | Emerging Markets Equity | LONG       |
| 6  | AAXJ   | 0.0361 | iShares MSCI All Country Asia ex Japan ... | Asia ex-Japan Equity    | LONG       |
| 18 | FNDE   | 0.0359 | Schwab Fundamental Emerging Markets Equ... | Emerging Markets Equity | LONG       |
| 12 | EEMA   | 0.0358 | iShares MSCI Emerging Markets Asia ETF     | Emerging Markets Equity | LONG       |
| 13 | PXH    | 0.0358 | Invesco RAFI Emerging Markets ETF          | Emerging Markets Equity | LONG       |
| 15 | GMF    | 0.0356 | State Street SPDR S&P Emerging Asia Pac... | Asia Equity             | LONG       |
| 9  | DGS    | 0.0354 | WisdomTree Emerging Market SmallCap Fund   | Emerging Markets Equity | LONG       |
| 20 | EEMV   | 0.0352 | iShares MSCI Emerging Markets Min Vol F... | Emerging Markets Equity | LONG       |
| 8  | DEM    | 0.0350 | WisdomTree Emerging Markets High Divide... | Emerging Markets Equity | LONG       |
| 11 | AIA    | 0.0348 | iShares Asia 50 ETF                        | Asia Equity             | LONG       |
| 5  | FEM    | 0.0342 | First Trust Emerging Markets AlphaDEX F... | Emerging Markets Equity | LONG       |
| 7  | EWX    | 0.0342 | State Street SPDR S&P Emerging Markets ... | Emerging Markets Equity | LONG       |
| 14 | DVYE   | 0.0322 | iShares Emerging Markets Dividend Index... | Emerging Markets Equity | LONG       |
| 10 | EWI    | 0.0313 | iShares MSCI Taiwan ETF                    | Taiwan Equity           | LONG       |
| 30 | FEMS   | 0.0308 | First Trust Emerging Markets Small Cap ... | Emerging Markets Equity | LONG       |
| 26 | SDIV   | 0.0299 | Global X SuperDividend ETF                 | Global Dividend Equity  | LONG       |
| 29 | EZA    | 0.0297 | iShares MSCI South Africa Index Fund       | South Africa Equity     | LONG       |
| 2  | EWY    | 0.0294 | iShares MSCI South Korea ETF               | South Korea Equity      | FLAT       |
| 28 | ILF    | 0.0275 | iShares Latin America 40 ETF               | Latin America Equity    | LONG       |
| 25 | EWI    | 0.0245 | iShares MSCI Mexico ETF                    | Mexico Equity           | LONG       |
| 1  | EWZ    | 0.0233 | iShares MSCI Brazil ETF                    | Brazil Equity           | LONG       |
| 27 | ECH    | 0.0226 | iShares MSCI Chile ETF                     | Chile Equity            | LONG       |

**Cluster 8 – U.S. Dividend Equity****Members:** 25   **Var explained:** 0.868   **Signal:** FLAT**Signal logic:**

- Latest buy\_cross: 2026-04-20 – filter **PASS** (252d at date = +20.8%)
- Latest sell\_cross: 2026-06-24
- Current 252d return: +17.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-20)

**Category breakdown**

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| U.S. Dividend Equity         | 13        | 0.5283     |
| U.S. Low Volatility Equity   | 4         | 0.1562     |
| Value Equity                 | 3         | 0.1200     |
| Dividend Equity              | 2         | 0.0792     |
| U.S. Defensive Equity        | 1         | 0.0404     |
| U.S. Quality / Factor Equity | 1         | 0.0390     |
| Global Equity                | 1         | 0.0368     |

**Replication basket** – SCHD (cor=0.96)   basket lead at lag 1: 0.004

| ticker | rep_wt | name                          | category             | etf_signal |
|--------|--------|-------------------------------|----------------------|------------|
| SCHD   | 0.891  | Schwab US Dividend Equity ETF | U.S. Dividend Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 7  | SDY    | 0.0418 | State Street SPDR S&P Dividend ETF         | U.S. Dividend Equity        | LONG       |
| 15 | DVY    | 0.0417 | iShares Select Dividend ETF                | U.S. Dividend Equity        | FLAT       |
| 2  | FVD    | 0.0415 | First Trust VL Dividend                    | U.S. Dividend Equity        | LONG       |
| 10 | SCHD   | 0.0413 | Schwab US Dividend Equity ETF              | U.S. Dividend Equity        | LONG       |
| 14 | SDOG   | 0.0413 | ALPS Sector Dividend Dogs ETF              | U.S. Dividend Equity        | FLAT       |
| 11 | FTA    | 0.0412 | First Trust Large Cap Value AlphaDEX Fu... | Value Equity                | LONG       |
| 20 | SPYD   | 0.0412 | State Street SPDR Portfolio S&P 500 Hig... | U.S. Dividend Equity        | FLAT       |
| 5  | DHS    | 0.0411 | WisdomTree U.S. High Dividend Fund         | U.S. Dividend Equity        | FLAT       |
| 19 | ONEY   | 0.0411 | State Street SPDR Russell 1000 Yield Fo... | U.S. Dividend Equity        | FLAT       |
| 18 | SPHD   | 0.0409 | Invesco S&P 500 High Dividend Low Volat... | Dividend Equity             | LONG       |
| 6  | CDC    | 0.0404 | VictoryShares US EQ Income Enhanced Vol... | U.S. Defensive Equity       | FLAT       |
| 4  | FDL    | 0.0402 | First Trust Morningstar ETF                | U.S. Dividend Equity        | LONG       |
| 25 | LGLV   | 0.0401 | State Street SPDR US Large Cap Low Vola... | U.S. Low Volatility Equi... | LONG       |
| 8  | LVHD   | 0.0399 | Franklin U.S. Low Volatility High Divid... | U.S. Low Volatility Equi... | FLAT       |
| 3  | PWV    | 0.0398 | Invesco Large Cap Value ETF                | Value Equity                | LONG       |
| 9  | HDV    | 0.0394 | iShares Core High Dividend ETF             | U.S. Dividend Equity        | LONG       |
| 16 | RDIV   | 0.0394 | Invesco S&P Ultra Dividend Revenue ETF     | U.S. Dividend Equity        | FLAT       |
| 12 | PEY    | 0.0393 | Invesco High Yield Equity Dividend Achi... | U.S. Dividend Equity        | FLAT       |
| 17 | RPV    | 0.0391 | Invesco S&P 500 Pure Value ETF             | Value Equity                | FLAT       |
| 13 | DIV    | 0.0390 | Global X Super Dividend ETF                | U.S. Dividend Equity        | FLAT       |
| 21 | FTCS   | 0.0390 | First Trust Capital Strength ETF           | U.S. Quality / Factor Eq... | LONG       |
| 22 | OUSA   | 0.0383 | ALPS O'Shares U.S. Quality Dividend ETF    | Dividend Equity             | LONG       |
| 23 | SPLV   | 0.0381 | Invesco S&P 500 Low Volatility ETF         | U.S. Low Volatility Equi... | FLAT       |
| 24 | USMV   | 0.0381 | iShares MSCI USA Min Vol Factor ETF        | U.S. Low Volatility Equi... | FLAT       |
| 1  | ACWV   | 0.0368 | iShares MSCI Global Min Vol Factor ETF     | Global Equity               | LONG       |

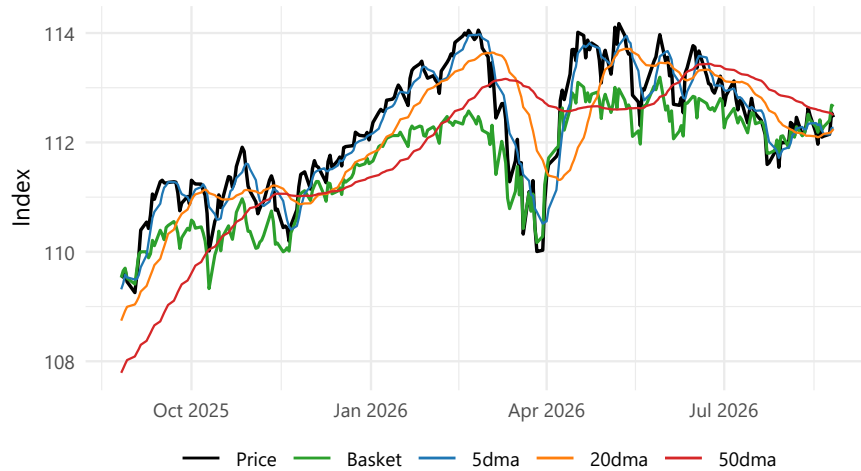
## Cluster 9 – High Yield Bond

**Members:** 23   **Var explained:** 0.738   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-06-17 – filter **PASS** (252d at date = +7.2%)

- Latest sell\_cross: 2026-07-07
- Current 252d return: +2.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| High Yield Bond               | 10        | 0.4783     |
| Preferred Stock               | 7         | 0.2821     |
| Emerging Markets Fixed Income | 6         | 0.2396     |

**Replication basket** – HYG (cor=0.94) basket lead at lag 1: 0.045

| ticker | rep_wt | name                                      | category        | etf_signal |
|--------|--------|-------------------------------------------|-----------------|------------|
| HYG    | 0.906  | iShares iBoxx \$ High Yield Corporate ... | High Yield Bond | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                        | category        | etf_signal |
|---|--------|--------|---------------------------------------------|-----------------|------------|
| 1 | HYG    | 0.0488 | iShares iBoxx \$ High Yield Corporate Bo... | High Yield Bond | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 13 | JNK  | 0.0488 | State Street SPDR Bloomberg High Yield ... | High Yield Bond             | LONG |
| 19 | HVLB | 0.0486 | Xtrackers USD High Yield Corporate Bond... | High Yield Bond             | LONG |
| 18 | BBHY | 0.0485 | JPMorgan BetaBuilders USD High Yield Co... | High Yield Bond             | LONG |
| 16 | SHYG | 0.0483 | iShares 0-5 Year High Yield Corporate B... | High Yield Bond             | LONG |
| 14 | SJNK | 0.0482 | State Street SPDR Bloomberg Short Term ... | High Yield Bond             | FLAT |
| 17 | FALN | 0.0480 | iShares Fallen Angels USD Bond ETF         | High Yield Bond             | FLAT |
| 15 | ANGL | 0.0479 | VanEck Fallen Angel High Yield Bond ETF    | High Yield Bond             | LONG |
| 6  | HYS  | 0.0474 | PIMCO 0-5 Year High Yield Corporat Bond... | High Yield Bond             | FLAT |
| 9  | PFF  | 0.0443 | iShares Preferred and Income Securities... | Preferred Stock             | LONG |
| 2  | EMB  | 0.0442 | iShares J.P. Morgan USD Emerging Market... | Emerging Markets Fixed I... | FLAT |
| 23 | VWOB | 0.0440 | Vanguard Emerging Markets Government Bo... | Emerging Markets Fixed I... | FLAT |
| 3  | HYLS | 0.0438 | First Trust Tactical High Yield ETF        | High Yield Bond             | FLAT |
| 20 | PCY  | 0.0432 | Invesco Emerging Markets Sovereign Debt... | Emerging Markets Fixed I... | FLAT |
| 11 | PFXF | 0.0424 | VanEck Preferred Securities ex Financia... | Preferred Stock             | LONG |
| 7  | PGX  | 0.0412 | Invesco Preferred ETF                      | Preferred Stock             | FLAT |
| 10 | PSK  | 0.0407 | State Street SPDR ICE Preferred Securit... | Preferred Stock             | FLAT |
| 4  | FPE  | 0.0400 | First Trust Preferred Securities and In... | Preferred Stock             | FLAT |
| 5  | PGF  | 0.0397 | Invesco Financial Preferred ETF            | Preferred Stock             | FLAT |
| 12 | HYEM | 0.0369 | VanEck Emerging Markets High Yield Bond... | Emerging Markets Fixed I... | LONG |
| 21 | EBND | 0.0365 | State Street SPDR Bloomberg Emerging Ma... | Emerging Markets Fixed I... | LONG |
| 22 | EMLC | 0.0348 | VanEck J. P. Morgan EM Local Currency B... | Emerging Markets Fixed I... | LONG |
| 8  | VRP  | 0.0339 | Invesco Variable Rate Preferred ETF        | Preferred Stock             | FLAT |

## Cluster 10 – Healthcare

**Members:** 17   **Var explained:** 0.756   **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell\_cross: 2026-07-23
- Current 252d return: +34.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



### Category breakdown

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Healthcare                | 16        | 0.9419     |
| U.S. Small Cap Healthcare | 1         | 0.0581     |

**Replication basket** – VHT (cor=0.93) basket lead at lag 1: -0.006

| ticker | rep_wt | name                     | category   | etf_signal |
|--------|--------|--------------------------|------------|------------|
| VHT    | 1.088  | Vanguard Health Care ETF | Healthcare | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category   | etf_signal |
|----|--------|--------|--------------------------------------------|------------|------------|
| 15 | VHT    | 0.0656 | Vanguard Health Care ETF                   | Healthcare | LONG       |
| 16 | FHLC   | 0.0655 | Fidelity MSCI Health Care Index ETF        | Healthcare | LONG       |
| 9  | FXH    | 0.0648 | First Trust Health Care AlphaDEX           | Healthcare | FLAT       |
| 14 | RSPH   | 0.0641 | Invesco S&P 500 Equal Weight Health Car... | Healthcare | FLAT       |
| 13 | IYH    | 0.0640 | iShares U.S. Healthcare ETF                | Healthcare | LONG       |
| 10 | IXJ    | 0.0625 | iShares Global Healthcare ETF              | Healthcare | FLAT       |

|    |      |        |                                            |                           |      |
|----|------|--------|--------------------------------------------|---------------------------|------|
| 1  | XLV  | 0.0623 | State Street Health Care Select Sector ... | Healthcare                | LONG |
| 3  | IBB  | 0.0605 | iShares Biotechnology ETF                  | Healthcare                | FLAT |
| 6  | FBT  | 0.0597 | First Trust Amex Biotech Index Fund        | Healthcare                | FLAT |
| 8  | PSCH | 0.0581 | Invesco S&P SmallCap Health Care ETF       | U.S. Small Cap Healthcare | FLAT |
| 7  | XHE  | 0.0571 | State Street SPDR S&P Health Care Equip... | Healthcare                | FLAT |
| 12 | IHI  | 0.0565 | iShares U.S. Medical Devices ETF           | Healthcare                | FLAT |
| 17 | PPH  | 0.0546 | VanEck Pharmaceutical ETF                  | Healthcare                | LONG |
| 5  | PTH  | 0.0537 | Invesco Dorsey Wright Healthcare Moment... | Healthcare                | LONG |
| 4  | XBI  | 0.0533 | State Street SPDR S&P Biotech ETF          | Healthcare                | FLAT |
| 2  | ARKG | 0.0507 | ARK Genomic Revolution ETF                 | Healthcare                | FLAT |
| 11 | IHF  | 0.0469 | iShares U.S. Health Care Providers ETF     | Healthcare                | FLAT |

### Cluster 11 – Materials

**Members:** 17   **Var explained:** 0.682   **Signal:** LONG since 2026-08-05 (15 days, 5.5%)

#### Signal logic:

- Latest buy\_cross: 2026-08-05 – filter **PASS** (252d at date = +34.1%)
- Latest sell\_cross: 2026-06-09
- Current 252d return: +36.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-06-09)



## Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| Materials                   | 9         | 0.5601     |
| MLP / Energy Infrastructure | 4         | 0.2162     |
| Energy                      | 1         | 0.0646     |
| Consumer Staples            | 1         | 0.0598     |
| Norway Equity               | 1         | 0.0557     |
| Industrial Metals           | 1         | 0.0436     |

**Replication basket** – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.046

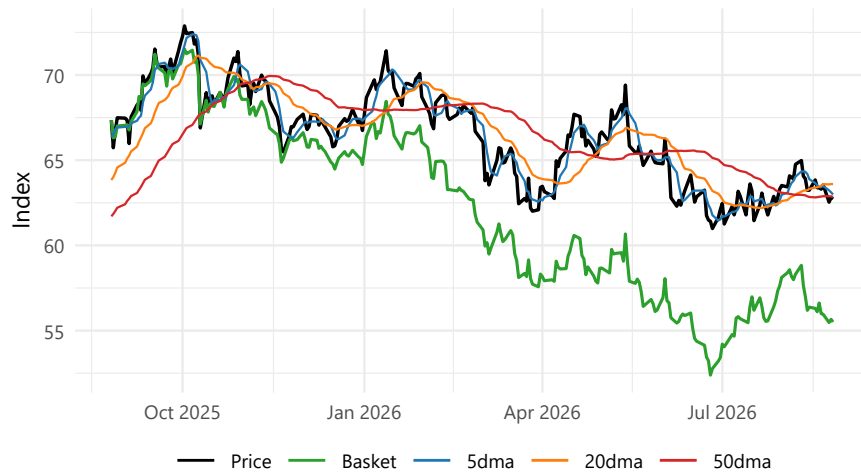
| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| AMLP   | 0.139  | Alerian MLP ETF                          | MLP / Energy Infrastr... | LONG       |
| URA    | 0.096  | Global X Uranium ETF                     | Materials                | LONG       |
| GUNR   | 0.772  | FlexShares Global Upstream Natural Re... | Materials                | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 11 | GNR    | 0.0694 | State Street SPDR S&P Global Natural Re... | Materials                   | LONG       |
| 10 | GUNR   | 0.0690 | FlexShares Global Upstream Natural Reso... | Materials                   | LONG       |
| 17 | NANR   | 0.0670 | State Street SPDR S&P North American Na... | Materials                   | LONG       |
| 16 | IGE    | 0.0646 | iShares North American Natural Resource... | Energy                      | LONG       |
| 9  | MXI    | 0.0635 | iShares Global Materials ETF               | Materials                   | LONG       |
| 12 | PICK   | 0.0633 | iShares MSCI Global Select Metals & Min... | Materials                   | LONG       |
| 3  | XME    | 0.0613 | State Street SPDR S&P Metals & Mining E... | Materials                   | LONG       |
| 13 | SLX    | 0.0609 | VanEck Steel ETF                           | Materials                   | FLAT       |
| 5  | MOO    | 0.0598 | VanEck Agribusiness ETF                    | Consumer Staples            | FLAT       |
| 2  | COPX   | 0.0591 | Global X Copper Miners ETF                 | Materials                   | LONG       |
| 8  | MLPX   | 0.0560 | Global X MLP & Energy Infrastructure ETF   | MLP / Energy Infrastruct... | LONG       |
| 14 | NORW   | 0.0557 | Global X MSCI Norway ETF                   | Norway Equity               | LONG       |
| 4  | EMLP   | 0.0545 | First Trust North American Energy Infra... | MLP / Energy Infrastruct... | LONG       |
| 6  | AMLP   | 0.0535 | Alerian MLP ETF                            | MLP / Energy Infrastruct... | LONG       |
| 7  | MLPA   | 0.0522 | Global X MLP ETF                           | MLP / Energy Infrastruct... | LONG       |
| 1  | URA    | 0.0467 | Global X Uranium ETF                       | Materials                   | LONG       |
| 15 | DBB    | 0.0436 | Invesco DB Base Metals Fund                | Industrial Metals           | LONG       |

**Cluster 12 – China Equity****Members:** 15   **Var explained:** 0.794   **Signal:** FLAT**Signal logic:**

- Latest buy\_cross: 2026-08-03 – filter **PASS** (252d at date = +2.6%)
- Latest sell\_cross: 2026-08-24
- Current 252d return: -6.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-24 more recent than last effective buy 2026-08-03)

**Category breakdown**

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| China Equity            | 13        | 0.8766     |
| Emerging Markets Equity | 1         | 0.0644     |
| Hong Kong Equity        | 1         | 0.0590     |

**Replication basket** – KWEB (cor=0.92)   basket lead at lag 1: 0.011

| ticker | rep_wt | name                               | category     | etf_signal |
|--------|--------|------------------------------------|--------------|------------|
| KWEB   | 0.561  | KraneShares CSI China Internet ETF | China Equity | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 11 | CXSE   | 0.0741 | WisdomTree China ex-State-Owned Enterpr... | China Equity            | FLAT       |
| 9  | GXC    | 0.0738 | State Street SPDR S&P China ETF            | China Equity            | FLAT       |
| 1  | MCHI   | 0.0732 | iShares MSCI China ETF                     | China Equity            | FLAT       |
| 2  | FXI    | 0.0712 | iShares China Large-Cap ETF                | China Equity            | FLAT       |
| 5  | CQQQ   | 0.0712 | Invesco China Technology ETF               | China Equity            | FLAT       |
| 7  | CHIQ   | 0.0705 | Global X MSCI China Consumer Discretion... | China Equity            | FLAT       |
| 4  | KWEB   | 0.0680 | KraneShares CSI China Internet ETF         | China Equity            | FLAT       |
| 6  | ASHR   | 0.0668 | Xtrackers Harvest CSI 300 China A-Share... | China Equity            | FLAT       |
| 8  | CNYA   | 0.0660 | iShares MSCI China A ETF                   | China Equity            | FLAT       |
| 13 | KGRN   | 0.0653 | KraneShares MSCI China Clean Technology... | China Equity            | FLAT       |
| 3  | EMQQ   | 0.0644 | EMQQ The Emerging Markets Internet ETF     | Emerging Markets Equity | FLAT       |
| 10 | KBA    | 0.0597 | KraneShares Boseri MSCI China A 50 Conn... | China Equity            | FLAT       |
| 12 | EWI    | 0.0590 | iShares MSCI Hong Kong Index Fund          | Hong Kong Equity        | FLAT       |
| 14 | CNXT   | 0.0587 | VanEck ChiNext Innovators ETF              | China Equity            | FLAT       |
| 15 | KURE   | 0.0581 | KraneShares MSCI All China Health Care ... | China Equity            | LONG       |

### Cluster 13 – Financials

**Members:** 14   **Var explained:** 0.871   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell\_cross: 2026-05-11
- Current 252d return: +11.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



### Category breakdown

| category      | n_members | abs_weight |
|---------------|-----------|------------|
| Financials    | 13        | 0.9383     |
| Global Equity | 1         | 0.0617     |

**Replication basket** – XLF (cor=0.96) basket lead at lag 1: 0.012

| ticker | rep_wt | name                                     | category   | etf_signal |
|--------|--------|------------------------------------------|------------|------------|
| XLF    | 1.094  | State Street Financial Select Sector ... | Financials | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category   | etf_signal |
|----|--------|--------|--------------------------------------------|------------|------------|
| 10 | VFH    | 0.0757 | Vanguard Financials ETF                    | Financials | FLAT       |
| 11 | FNCL   | 0.0756 | Fidelity MSCI Financials Index ETF         | Financials | LONG       |
| 7  | IYF    | 0.0754 | iShares U.S. Financial ETF                 | Financials | FLAT       |
| 5  | FXO    | 0.0752 | First Trust Financials AlphaDEX            | Financials | FLAT       |
| 8  | IYG    | 0.0749 | iShares U.S. Financial Services ETF        | Financials | FLAT       |
| 1  | XLF    | 0.0742 | State Street Financial Select Sector SP... | Financials | LONG       |

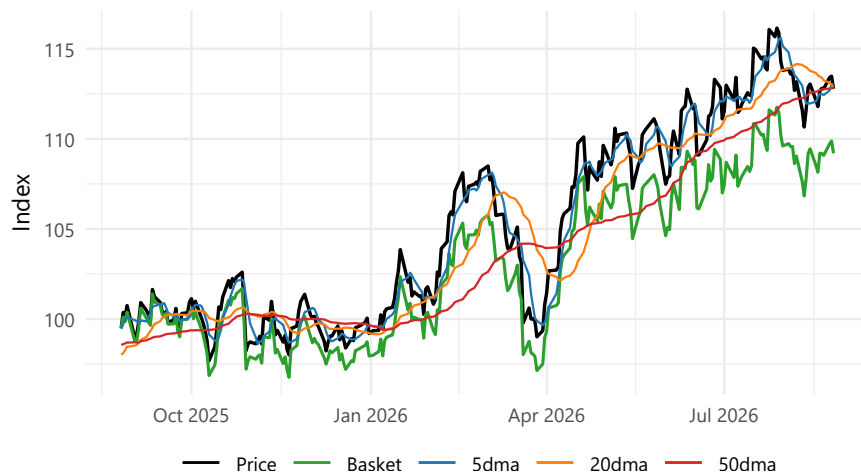
|    |      |        |                                            |               |      |
|----|------|--------|--------------------------------------------|---------------|------|
| 9  | KBWB | 0.0733 | Invesco KBW Bank ETF                       | Financials    | FLAT |
| 3  | KBE  | 0.0713 | State Street SPDR S&P Bank ETF             | Financials    | FLAT |
| 14 | KCE  | 0.0707 | State Street SPDR S&P Capital Markets E... | Financials    | LONG |
| 6  | IAT  | 0.0706 | iShares U.S. Regional Banks ETF            | Financials    | FLAT |
| 13 | IAI  | 0.0695 | iShares U.S. Broker-Dealers & Securitie... | Financials    | FLAT |
| 2  | KRE  | 0.0687 | State Street SPDR S&P Regional Banking ... | Financials    | FLAT |
| 4  | KIE  | 0.0630 | State Street SPDR S&P Insurance ETF        | Financials    | FLAT |
| 12 | PSP  | 0.0617 | Invesco Global Listed Private Equity ETF   | Global Equity | FLAT |

### Cluster 14 – Real Estate

**Members:** 14   **Var explained:** 0.961   **Signal:** LONG since 2026-08-25 (1 days, -0.6%)

#### Signal logic:

- Latest buy\_cross: 2026-08-25 – filter **PASS** (252d at date = +13.1%)
- Latest sell\_cross: 2026-08-06
- Current 252d return: +13.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-25 > last sell at 2026-08-06)



### Category breakdown

| category    | n_members | abs_weight |
|-------------|-----------|------------|
| Real Estate | 14        | 1          |

**Replication basket** – VNQ (cor=0.99) basket lead at lag 1: -0.001

| ticker | rep_wt | name                     | category    | etf_signal |
|--------|--------|--------------------------|-------------|------------|
| VNQ    | 0.968  | Vanguard Real Estate ETF | Real Estate | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category    | etf_signal |
|----|--------|--------|--------------------------------------------|-------------|------------|
| 8  | VNQ    | 0.0725 | Vanguard Real Estate ETF                   | Real Estate | LONG       |
| 9  | FREL   | 0.0725 | Fidelity MSCI Real Estate Index ETF        | Real Estate | LONG       |
| 11 | SCHH   | 0.0725 | Schwab U.S. REIT ETF                       | Real Estate | LONG       |
| 2  | IYR    | 0.0724 | iShares U.S. Real Estate ETF               | Real Estate | FLAT       |
| 10 | USRT   | 0.0724 | iShares Core U.S. REIT ETF                 | Real Estate | FLAT       |
| 6  | RWR    | 0.0723 | State Street SPDR Dow Jones REIT ETF       | Real Estate | LONG       |
| 13 | PSR    | 0.0721 | Invesco Active U.S. Real Estate Fund       | Real Estate | FLAT       |
| 1  | XLRE   | 0.0720 | State Street Real Estate Select Sector ... | Real Estate | LONG       |
| 5  | ICF    | 0.0720 | iShares Select U.S. REIT ETF               | Real Estate | LONG       |
| 7  | REET   | 0.0716 | iShares Global REIT ETF                    | Real Estate | LONG       |
| 3  | RWO    | 0.0712 | State Street SPDR Dow Jones Global Real... | Real Estate | LONG       |
| 4  | REZ    | 0.0696 | iShares Residential and Multisector Rea... | Real Estate | LONG       |
| 14 | NURE   | 0.0687 | Nuveen Short-Term REIT ETF                 | Real Estate | FLAT       |
| 12 | INDS   | 0.0682 | Pacer Industrial Real Estate ETF           | Real Estate | FLAT       |

## Cluster 15 – Technology

**Members:** 14 **Var explained:** 0.827 **Signal:** LONG since 2026-08-04 (16 days, 1.2%)

### Signal logic:

- Latest buy\_cross: 2026-08-04 – filter **PASS** (252d at date = +6.7%)
- Latest sell\_cross: 2026-07-23
- Current 252d return: +4.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-04 > last sell at 2026-07-23)



### Category breakdown

| category               | n_members | abs_weight |
|------------------------|-----------|------------|
| Technology             | 10        | 0.7125     |
| Financials             | 2         | 0.1429     |
| IPO Equity             | 1         | 0.0731     |
| Consumer Discretionary | 1         | 0.0715     |

**Replication basket** – IGV (cor=0.90) basket lead at lag 1: 0.054

| ticker | rep_wt | name                                     | category   | etf_signal |
|--------|--------|------------------------------------------|------------|------------|
| IGV    | 0.951  | iShares Expanded Tech-Software Sector... | Technology | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                               | category   | etf_signal |
|----|--------|--------|------------------------------------|------------|------------|
| 12 | FDN    | 0.0747 | First Trust DJ Internet Index Fund | Technology | LONG       |
| 6  | ARKW   | 0.0745 | ARK Next Generation Internet ETF   | Technology | LONG       |
| 2  | SKYY   | 0.0741 | First Trust Cloud Computing ETF    | Technology | LONG       |
| 14 | FINX   | 0.0740 | Global X FinTech ETF               | Financials | FLAT       |

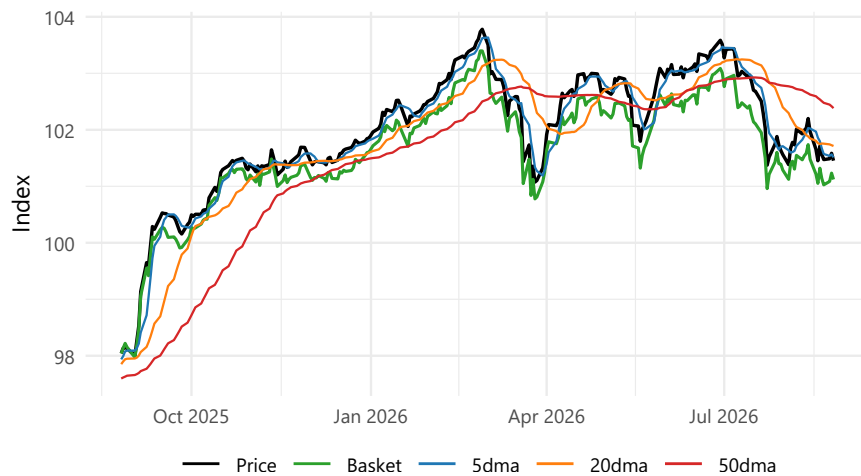
|    |      |        |                                            |                        |      |
|----|------|--------|--------------------------------------------|------------------------|------|
| 11 | PNQI | 0.0734 | Invesco Nasdaq Internet ETF                | Technology             | FLAT |
| 10 | IPO  | 0.0731 | Renaissance IPO ETF                        | IPO Equity             | LONG |
| 5  | ARKK | 0.0730 | ARK Innovation ETF                         | Technology             | LONG |
| 1  | IGV  | 0.0720 | iShares Expanded Tech-Software Sector E... | Technology             | FLAT |
| 8  | IBUY | 0.0715 | Amplify Online Retail ETF                  | Consumer Discretionary | FLAT |
| 9  | CIBR | 0.0693 | First Trust NASDAQ Cybersecurity ETF       | Technology             | FLAT |
| 7  | ARKQ | 0.0691 | ARK Autonomous Technology & Robotics ETF   | Technology             | LONG |
| 13 | IPAY | 0.0689 | Amplify Digital Payments ETF               | Financials             | FLAT |
| 4  | HACK | 0.0683 | Amplify Cybersecurity ETF                  | Technology             | FLAT |
| 3  | BLOK | 0.0641 | Amplify Blockchain Technology ETF          | Technology             | LONG |

### Cluster 16 – Municipal Bond

**Members:** 13   **Var explained:** 0.820   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell\_cross: 2026-08-21
- Current 252d return: +3.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-21 more recent than last effective buy 2026-05-28)



**Category breakdown**

| category       | n_members | abs_weight |
|----------------|-----------|------------|
| Municipal Bond | 13        | 1          |

**Replication basket** – MUB (cor=0.95)    basket lead at lag 1: 0.178

| ticker | rep_wt | name                           | category       | etf_signal |
|--------|--------|--------------------------------|----------------|------------|
| MUB    | 0.904  | iShares National Muni Bond ETF | Municipal Bond | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category       | etf_signal |
|----|--------|--------|--------------------------------------------|----------------|------------|
| 13 | VTEB   | 0.0817 | Vanguard Tax-Exempt Bond ETF               | Municipal Bond | FLAT       |
| 11 | MUB    | 0.0813 | iShares National Muni Bond ETF             | Municipal Bond | FLAT       |
| 6  | CMF    | 0.0802 | iShares California Muni Bond ETF           | Municipal Bond | FLAT       |
| 9  | TFI    | 0.0801 | State Street SPDR Nuveen ICE Municipal ... | Municipal Bond | FLAT       |
| 8  | ITM    | 0.0800 | VanEck Intermediate Muni ETF               | Municipal Bond | FLAT       |
| 7  | NYF    | 0.0794 | iShares New York Muni Bond ETF             | Municipal Bond | FLAT       |
| 1  | FMB    | 0.0793 | First Trust Managed Municipal ETF          | Municipal Bond | FLAT       |
| 5  | PZA    | 0.0780 | Invesco National AMT-Free Municipal Bon... | Municipal Bond | FLAT       |
| 4  | MUNI   | 0.0770 | PIMCO Intermediate Municipal Bond Activ... | Municipal Bond | FLAT       |
| 2  | HYD    | 0.0748 | VanEck High Yield Muni ETF                 | Municipal Bond | FLAT       |
| 3  | HYMB   | 0.0722 | State Street SPDR Nuveen ICE High Yield... | Municipal Bond | FLAT       |
| 12 | SUB    | 0.0693 | iShares Short-Term National Muni Bond E... | Municipal Bond | LONG       |
| 10 | SHM    | 0.0667 | State Street SPDR Nuveen ICE Short Term... | Municipal Bond | FLAT       |

**Cluster 17 – Gold**

**Members:** 12    **Var explained:** 0.864    **Signal:** LONG since 2026-08-07 (13 days, 8.6%)

**Signal logic:**

- Latest buy\_cross: 2026-08-07 – filter **PASS** (252d at date = +36.2%)
- Latest sell\_cross: 2026-05-20
- Current 252d return: +44.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-05-20)



### Category breakdown

| category        | n_members | abs_weight |
|-----------------|-----------|------------|
| Materials       | 5         | 0.4158     |
| Gold            | 3         | 0.2503     |
| Precious Metals | 2         | 0.1723     |
| Silver          | 2         | 0.1616     |

**Replication basket** – GLD+IAU+SLV (cor=0.95)    basket lead at lag 1: -0.014

| ticker | rep_wt | name                      | category | etf_signal |
|--------|--------|---------------------------|----------|------------|
| GLD    | 0.024  | SPDR Gold Shares          | Gold     | LONG       |
| IAU    | 0.754  | iShares Gold Trust Shares | Gold     | LONG       |
| SLV    | 0.411  | iShares Silver Trust      | Silver   | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category        | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------|------------|
| 11 | DBP    | 0.0865 | Invesco DB Precious Metals Fund            | Precious Metals | LONG       |
| 7  | GLTR   | 0.0858 | abrdn Physical Precious Metals Basket S... | Precious Metals | LONG       |

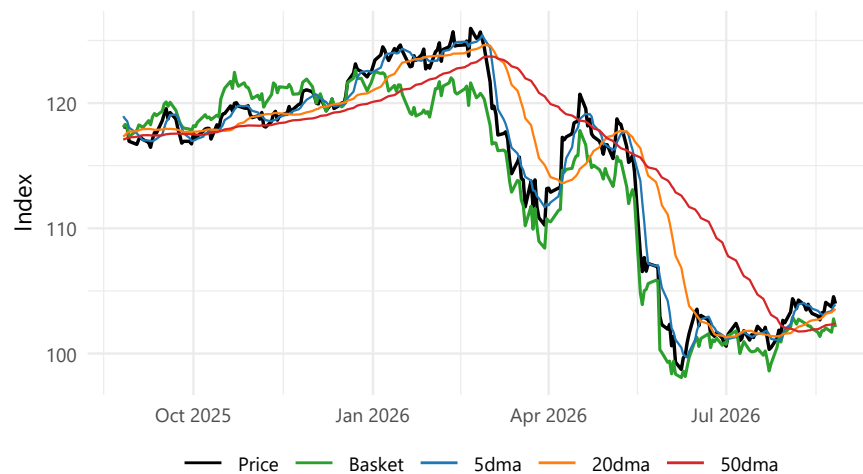
|    |      |        |                                            |           |      |
|----|------|--------|--------------------------------------------|-----------|------|
| 2  | GDXJ | 0.0841 | VanEck Junior Gold Miners ETF              | Materials | LONG |
| 1  | GDX  | 0.0840 | VanEck Gold Miners ETF                     | Materials | LONG |
| 3  | GLD  | 0.0835 | SPDR Gold Shares                           | Gold      | LONG |
| 9  | IAU  | 0.0835 | iShares Gold Trust Shares                  | Gold      | LONG |
| 10 | SGOL | 0.0833 | abrtn Physical Gold Shares ETF             | Gold      | LONG |
| 12 | GOAU | 0.0831 | US Global GO Gold and Precious Metal Mi... | Materials | LONG |
| 6  | SIL  | 0.0829 | Global X Silver Miners ETF                 | Materials | LONG |
| 5  | SILJ | 0.0818 | Amplify Junior Silver Miners ETF           | Materials | LONG |
| 4  | SLV  | 0.0809 | iShares Silver Trust                       | Silver    | LONG |
| 8  | SIVR | 0.0807 | abrtn Physical Silver Shares ETF           | Silver    | LONG |

### Cluster 18 – India Equity

**Members:** 12   **Var explained:** 0.339   **Signal:** FLAT

#### Signal logic:

- Latest buy\_cross: 2026-08-03 – filter **BLOCKED** (252d at date = -9.8%; needs > 0)
- Latest effective buy: 2026-05-06 – delayed re-arm from blocked cross (252d at fire = +4.6%)
- Latest sell\_cross: 2026-07-28
- Current 252d return: -12.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-28 more recent than last effective buy 2026-05-06)



## Category breakdown

| category                 | n_members | abs_weight |
|--------------------------|-----------|------------|
| India Equity             | 3         | 0.4526     |
| Malaysia Equity          | 1         | 0.1094     |
| Indonesia Equity         | 1         | 0.1012     |
| Philippines Equity       | 1         | 0.0981     |
| Vietnam Equity           | 1         | 0.0730     |
| Turkey Equity            | 1         | 0.0589     |
| U.S. Dividend Equity     | 1         | 0.0530     |
| Agricultural Commodities | 1         | 0.0254     |
| Platinum                 | 1         | 0.0232     |
| Natural Gas              | 1         | 0.0053     |

**Replication basket** – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.005

| ticker | rep_wt | name                                     | category             | etf_signal |
|--------|--------|------------------------------------------|----------------------|------------|
| NOBL   | 0.078  | ProShares S&P 500 Dividend Aristocrat... | U.S. Dividend Equity | FLAT       |
| INDA   | 0.641  | iShares MSCI India ETF                   | India Equity         | FLAT       |
| PPLT   | 0.029  | abrdn Physical Platinum Shares ETF       | Platinum             | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                 | etf_signal |
|----|--------|--------|--------------------------------------------|--------------------------|------------|
| 1  | INDA   | 0.1522 | iShares MSCI India ETF                     | India Equity             | FLAT       |
| 3  | INDY   | 0.1505 | iShares India 50 ETF                       | India Equity             | FLAT       |
| 4  | EPI    | 0.1500 | WisdomTree India Earnings Fund             | India Equity             | FLAT       |
| 10 | EWM    | 0.1094 | iShares MSCI Malaysia Index Fund           | Malaysia Equity          | LONG       |
| 8  | EIDO   | 0.1012 | iShares MSCI Indonesia ETF                 | Indonesia Equity         | FLAT       |
| 9  | EPHE   | 0.0981 | iShares MSCI Philippines ETF               | Philippines Equity       | FLAT       |
| 12 | VNM    | 0.0730 | VanEck Vietnam ETF                         | Vietnam Equity           | LONG       |
| 11 | TUR    | 0.0589 | iShares MSCI Turkey ETF                    | Turkey Equity            | LONG       |
| 5  | NOBL   | 0.0530 | ProShares S&P 500 Dividend Aristocrats ... | U.S. Dividend Equity     | FLAT       |
| 2  | DBA    | 0.0254 | Invesco DB Agriculture Fund                | Agricultural Commodities | FLAT       |
| 6  | PPLT   | 0.0232 | abrdn Physical Platinum Shares ETF         | Platinum                 | FLAT       |
| 7  | UNG    | 0.0053 | United States Natural Gas Fund LP          | Natural Gas              | FLAT       |

**Cluster 19 – Consumer Staples**

**Members:** 11   **Var explained:** 0.740   **Signal:** LONG since 2026-08-25 (1 days, -0.1%)

**Signal logic:**

- Latest buy\_cross: 2026-08-25 – filter **PASS** (252d at date = +4.2%)
- Latest sell\_cross: 2026-08-05
- Current 252d return: +5.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-25 > last sell at 2026-08-05)

**Category breakdown**

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Consumer Staples                | 6         | 0.5744     |
| Utilities                       | 4         | 0.3503     |
| U.S. Small Cap Consumer Staples | 1         | 0.0753     |

**Replication basket** – XLU+XLP (cor=0.99)   basket lead at lag 1: -0.017

| ticker | rep_wt | name                                     | category  | etf_signal |
|--------|--------|------------------------------------------|-----------|------------|
| XLU    | 0.363  | State Street Utilities Select Sector ... | Utilities | FLAT       |

XLP      0.587      State Street Consumer Staples Select ...      Consumer Staples      LONG

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|    | ticker | PC_wt  | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5  | VDC    | 0.0984 | Vanguard Consumer Staples ETF              | Consumer Staples            | LONG       |
| 6  | FSTA   | 0.0984 | Fidelity MSCI Consumer Staples Index ETF   | Consumer Staples            | LONG       |
| 1  | XLP    | 0.0976 | State Street Consumer Staples Select Se... | Consumer Staples            | LONG       |
| 4  | IYK    | 0.0951 | iShares U.S. Consumer Staples ETF          | Consumer Staples            | FLAT       |
| 3  | KXI    | 0.0948 | iShares Global Consumer Staples ETF        | Consumer Staples            | FLAT       |
| 11 | PBJ    | 0.0900 | Invesco Food & Beverage ETF                | Consumer Staples            | LONG       |
| 7  | IDU    | 0.0887 | iShares U.S. Utilities ETF                 | Utilities                   | FLAT       |
| 9  | FUTY   | 0.0874 | Fidelity MSCI Utilities Index ETF          | Utilities                   | FLAT       |
| 8  | VPU    | 0.0873 | Vanguard Utilities ETF                     | Utilities                   | FLAT       |
| 2  | XLU    | 0.0870 | State Street Utilities Select Sector SP... | Utilities                   | FLAT       |
| 10 | PSCC   | 0.0753 | Invesco S&P SmallCap Consumer Staples E... | U.S. Small Cap Consumer ... | LONG       |

## Cluster 20 – Inflation-Protected Fixed Income

**Members:** 10    **Var explained:** 0.904    **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell\_cross: 2026-07-02
- Current 252d return: -0.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



### Category breakdown

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Inflation-Protected Fixed Income | 10        | 1          |

**Replication basket** – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| VTIP   | 0.267  | Vanguard Short-Term Inflation-Protect... | Inflation-Protected F... | LONG       |
| SCHP   | 0.760  | Schwab U.S. TIPS ETF                     | Inflation-Protected F... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 4 | TDTF   | 0.1036 | FlexShares iBoxx 5 Year Target Duration... | Inflation-Protected Fixe... | FLAT       |
| 6 | TIPX   | 0.1035 | State Street SPDR Bloomberg 1a??10 Year... | Inflation-Protected Fixe... | FLAT       |
| 3 | TIP    | 0.1030 | iShares TIPS Bond ETF                      | Inflation-Protected Fixe... | FLAT       |
| 8 | SCHP   | 0.1024 | Schwab U.S. TIPS ETF                       | Inflation-Protected Fixe... | FLAT       |
| 5 | TDTT   | 0.1008 | FlexShares iBoxx 3 Year Target Duration... | Inflation-Protected Fixe... | FLAT       |
| 7 | SPIP   | 0.1000 | State Street SPDR Portfolio TIPS ETF       | Inflation-Protected Fixe... | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 9  | STIP | 0.0997 | iShares 0-5 Year TIPS Bond ETF             | Inflation-Protected Fixe... | FLAT |
| 2  | STPZ | 0.0996 | PIMCO 1-5 Year U.S. TIPS Index Exchange... | Inflation-Protected Fixe... | FLAT |
| 10 | VTIP | 0.0989 | Vanguard Short-Term Inflation-Protected... | Inflation-Protected Fixe... | LONG |
| 1  | LTPZ | 0.0884 | Pimco 15 Year U.S. TIPS Index Exchange...  | Inflation-Protected Fixe... | FLAT |

## Cluster 21 – Japan Equity

**Members:** 9   **Var explained:** 0.887   **Signal:** LONG since 2026-08-06 (14 days, 0.6%)

### Signal logic:

- Latest buy\_cross: 2026-08-06 – filter **PASS** (252d at date = +35.6%)
- Latest sell\_cross: 2026-07-28
- Current 252d return: +29.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-06 > last sell at 2026-07-28)



### Category breakdown

| category     | n_members | abs_weight |
|--------------|-----------|------------|
| Japan Equity | 7         | 0.7805     |

|                     |   |        |
|---------------------|---|--------|
| Asia-Pacific Equity | 2 | 0.2195 |
|---------------------|---|--------|

**Replication basket** – EWJ (cor=0.97) basket lead at lag 1: -0.043

| ticker | rep_wt | name                          | category     | etf_signal |
|--------|--------|-------------------------------|--------------|------------|
| EWJ    | 0.907  | iShares MSCI Japan Index Fund | Japan Equity | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

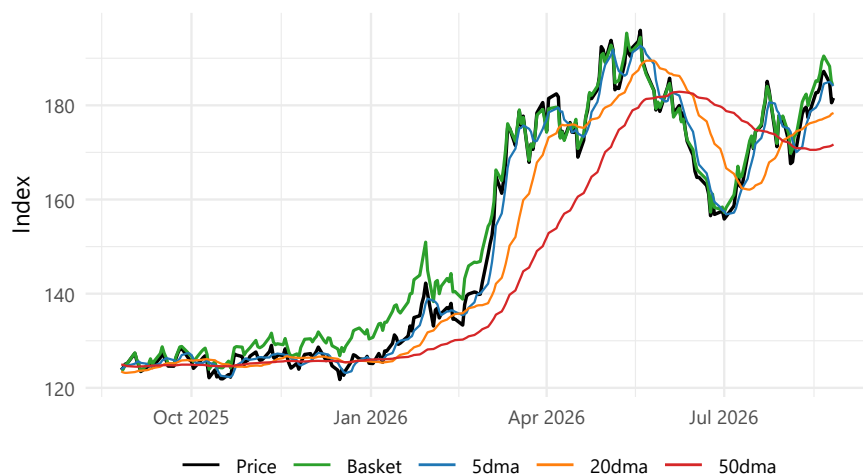
|   | ticker | PC_wt  | name                                   | category            | etf_signal |
|---|--------|--------|----------------------------------------|---------------------|------------|
| 1 | EWJ    | 0.1142 | iShares MSCI Japan Index Fund          | Japan Equity        | LONG       |
| 8 | BBJP   | 0.1142 | JPMorgan BetaBuilders Japan ETF        | Japan Equity        | LONG       |
| 6 | DBJP   | 0.1118 | Xtrackers MSCI Japan Hedged Equity ETF | Japan Equity        | LONG       |
| 9 | HEWJ   | 0.1118 | iShares Currency Hedged MSCI Japan ETF | Japan Equity        | LONG       |
| 3 | IPAC   | 0.1114 | iShares Core MSCI Pacific ETF          | Asia-Pacific Equity | LONG       |
| 5 | FLJH   | 0.1110 | Franklin FTSE Japan Hedged ETF         | Japan Equity        | LONG       |
| 2 | DXJ    | 0.1096 | WisdomTree Japan Hedged Equity Fund    | Japan Equity        | LONG       |
| 4 | VPL    | 0.1081 | Vanguard FTSE Pacific ETF              | Asia-Pacific Equity | LONG       |
| 7 | EWJV   | 0.1079 | iShares MSCI Japan Value ETF           | Japan Equity        | LONG       |

## Cluster 22 – Broad Commodities

**Members:** 9 **Var explained:** 0.911 **Signal:** LONG since 2026-08-10 (12 days, 1.9%)

### Signal logic:

- Latest buy\_cross: 2026-08-10 – filter **PASS** (252d at date = +45.4%)
- Latest sell\_cross: 2026-08-06
- Current 252d return: +45.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-10 > last sell at 2026-08-06)



### Category breakdown

| category           | n_members | abs_weight |
|--------------------|-----------|------------|
| Broad Commodities  | 6         | 0.6698     |
| Oil                | 2         | 0.2183     |
| Energy Commodities | 1         | 0.1119     |

**Replication basket** – PDBC (cor=0.98) basket lead at lag 1: 0.076

| ticker | rep_wt | name                                     | category          | etf_signal |
|--------|--------|------------------------------------------|-------------------|------------|
| PDBC   | 1.181  | Invesco Optimum Yield Diversified Com... | Broad Commodities | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category           | etf_signal |
|---|--------|--------|--------------------------------------------|--------------------|------------|
| 5 | GSG    | 0.1152 | iShares GSCI Commodity-Indexed Trust Fu... | Broad Commodities  | FLAT       |
| 2 | DBC    | 0.1151 | Invesco DB Commodity Index Tracking Fund   | Broad Commodities  | FLAT       |
| 7 | PDBC   | 0.1150 | Invesco Optimum Yield Diversified Commo... | Broad Commodities  | FLAT       |
| 8 | COMT   | 0.1145 | iShares GSCI Commodity Dynamic Roll Str... | Broad Commodities  | FLAT       |
| 9 | DBE    | 0.1119 | Invesco DB Energy Fund                     | Energy Commodities | FLAT       |

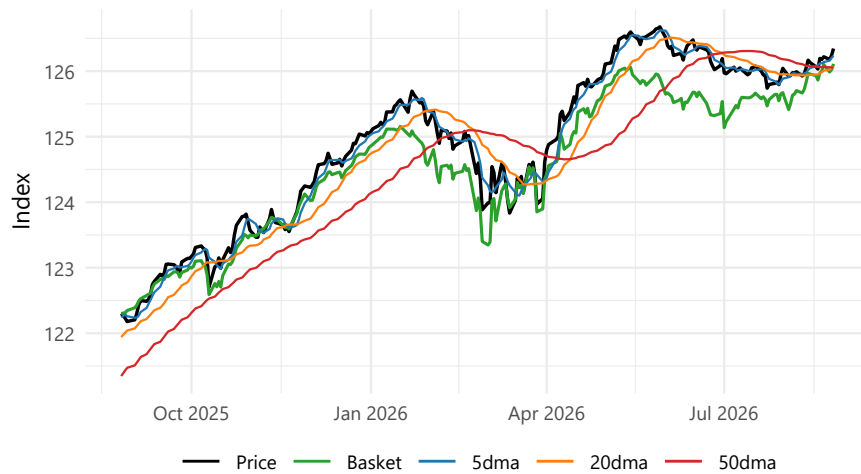
|   |      |        |                                            |                   |      |
|---|------|--------|--------------------------------------------|-------------------|------|
| 1 | USO  | 0.1098 | United States Oil Fund                     | Oil               | LONG |
| 4 | DBO  | 0.1084 | Invesco DB Oil Fund                        | Oil               | LONG |
| 6 | DJP  | 0.1055 | iPath Bloomberg Commodity Index Total R... | Broad Commodities | FLAT |
| 3 | FTGC | 0.1044 | First Trust Global Tactical Commodity S... | Broad Commodities | FLAT |

### Cluster 23 – Bank Loan / Floating Rate

**Members:** 9   **Var explained:** 0.555   **Signal:** LONG since 2026-08-17 (7 days, 0.2%)

#### Signal logic:

- Latest buy\_cross: 2026-08-17 – filter **PASS** (252d at date = +3.2%)
- Latest sell\_cross: 2026-07-23
- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-17 > last sell at 2026-07-23)



#### Category breakdown

| category                          | n_members | abs_weight |
|-----------------------------------|-----------|------------|
| Bank Loan / Floating Rate         | 3         | 0.3618     |
| Interest Rate Hedged Fixed Income | 3         | 0.3245     |

Floating Rate Fixed Income 3 0.3137

**Replication basket** – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| FLOT   | 0.294  | iShares Floating Rate Bond ETF           | Floating Rate Fixed I... | LONG       |
| BKLN   | 0.215  | Invesco Senior Loan ETF                  | Bank Loan / Floating ... | LONG       |
| SRLN   | 0.300  | State Street Blackstone Senior Loan E... | Bank Loan / Floating ... | LONG       |
| FLRN   | 0.123  | State Street SPDR Bloomberg Investmen... | Floating Rate Fixed I... | LONG       |
| FLTR   | 0.034  | VanEck IG Floating Rate ETF              | Floating Rate Fixed I... | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

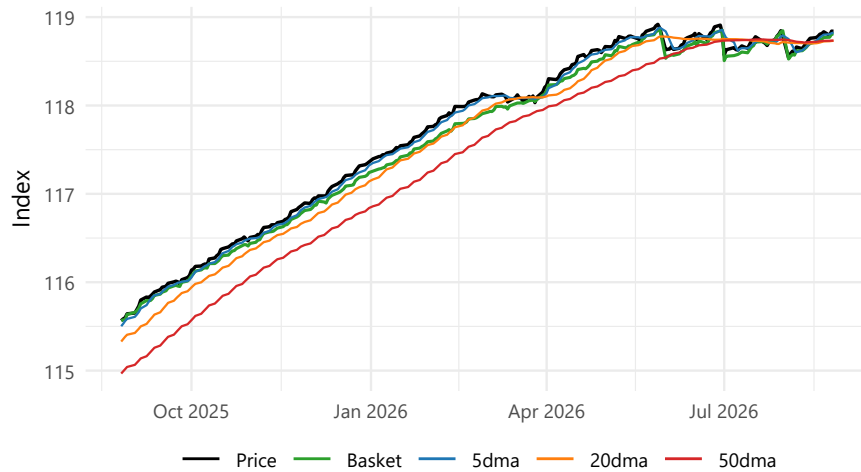
|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | SRLN   | 0.1251 | State Street Blackstone Senior Loan ETF    | Bank Loan / Floating Rate   | LONG       |
| 3 | BKLN   | 0.1207 | Invesco Senior Loan ETF                    | Bank Loan / Floating Rate   | LONG       |
| 1 | FTSL   | 0.1160 | First Trust Senior Loan Fund               | Bank Loan / Floating Rate   | FLAT       |
| 5 | LQDH   | 0.1143 | iShares Interest Rate Hedged Corporate ... | Interest Rate Hedged Fix... | FLAT       |
| 7 | FLOT   | 0.1085 | iShares Floating Rate Bond ETF             | Floating Rate Fixed Inco... | LONG       |
| 6 | IGBH   | 0.1056 | iShares Interest Rate Hedged Long-Term ... | Interest Rate Hedged Fix... | LONG       |
| 4 | IGHG   | 0.1047 | ProShares Investment Grade-Interest Rat... | Interest Rate Hedged Fix... | FLAT       |
| 8 | FLRN   | 0.1040 | State Street SPDR Bloomberg Investment ... | Floating Rate Fixed Inco... | LONG       |
| 9 | FLTR   | 0.1012 | VanEck IG Floating Rate ETF                | Floating Rate Fixed Inco... | LONG       |

## Cluster 24 – Cash / Ultra-Short Fixed Income

**Members:** 9 **Var explained:** 0.459 **Signal:** LONG since 2026-08-17 (7 days, 0.1%)

### Signal logic:

- Latest buy\_cross: 2026-08-17 – filter **PASS** (252d at date = +3.0%)
- Latest sell\_cross: 2026-08-05
- Current 252d return: +2.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-17 > last sell at 2026-08-05)



### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Cash / Ultra-Short Fixed Income | 9         | 1          |

**Replication basket** – SHV+MINT+GBIL+ICSH (cor=0.86) basket lead at lag 1: 0.046

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| SHV    | 0.254  | iShares 0-1 Year Treasury Bond ETF       | Cash / Ultra-Short Fi... | LONG       |
| MINT   | 0.174  | PIMCO Enhanced Short Maturity Active ... | Cash / Ultra-Short Fi... | LONG       |
| GBIL   | 0.126  | Goldman Sachs Access Treasury 0-1 Yea... | Cash / Ultra-Short Fi... | LONG       |
| ICSH   | 0.398  | iShares Ultra Short Duration Bond Act... | Cash / Ultra-Short Fi... | LONG       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 7 | SHV    | 0.1330 | iShares 0-1 Year Treasury Bond ETF         | Cash / Ultra-Short Fixed... | LONG       |
| 6 | ICSH   | 0.1311 | iShares Ultra Short Duration Bond Activ... | Cash / Ultra-Short Fixed... | LONG       |
| 2 | MINT   | 0.1160 | PIMCO Enhanced Short Maturity Active Ex... | Cash / Ultra-Short Fixed... | LONG       |
| 4 | NEAR   | 0.1101 | iShares Short Duration Bond Active ETF     | Cash / Ultra-Short Fixed... | LONG       |

|   |      |        |                                            |                             |      |
|---|------|--------|--------------------------------------------|-----------------------------|------|
| 5 | GSY  | 0.1100 | Invesco Ultra Short Duration ETF           | Cash / Ultra-Short Fixed... | LONG |
| 8 | BIL  | 0.1072 | State Street SPDR Bloomberg 1-3 Month T... | Cash / Ultra-Short Fixed... | LONG |
| 3 | FTSM | 0.1053 | First Trust Enhanced Short Maturity ETF    | Cash / Ultra-Short Fixed... | LONG |
| 9 | GBIL | 0.0953 | Goldman Sachs Access Treasury 0-1 Year ... | Cash / Ultra-Short Fixed... | LONG |
| 1 | VNLA | 0.0920 | Janus Henderson Short Duration Income E... | Cash / Ultra-Short Fixed... | LONG |

## Cluster 25 – International Fixed Income

**Members:** 6 **Var explained:** 0.742 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-08-04 – filter **BLOCKED** (252d at date = -1.4%; needs > 0)
- Latest effective buy: 2026-05-29 – delayed re-arm from blocked cross (252d at fire = +0.6%)
- Latest sell\_cross: 2026-07-23
- Current 252d return: -1.2% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



### Category breakdown

| category | n_members | abs_weight |
|----------|-----------|------------|
|----------|-----------|------------|

|                            |   |        |
|----------------------------|---|--------|
| Currency                   | 4 | 0.6409 |
| International Fixed Income | 2 | 0.3591 |

**Replication basket** – BWX (cor=0.93) basket lead at lag 1: 0.023

| ticker | rep_wt | name                                     | category                 | etf_signal |
|--------|--------|------------------------------------------|--------------------------|------------|
| BWX    | 0.752  | State Street SPDR Bloomberg Internati... | International Fixed I... | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

|   | ticker | PC_wt  | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | IGOV   | 0.1801 | iShares International Treasury Bond ETF    | International Fixed Inco... | LONG       |
| 1 | BWX    | 0.1790 | State Street SPDR Bloomberg Internation... | International Fixed Inco... | FLAT       |
| 4 | FXE    | 0.1702 | Invesco CurrencyShares Euro Currency Tr... | Currency                    | LONG       |
| 5 | FXF    | 0.1651 | Invesco CurrencyShares Swiss Franc Trust   | Currency                    | LONG       |
| 3 | FXB    | 0.1608 | Invesco CurrencyShares British Pound St... | Currency                    | LONG       |
| 6 | FXV    | 0.1448 | Invesco CurrencyShares Japanese Yen Tru... | Currency                    | FLAT       |

## Cluster 26 – Leveraged / Inverse

**Members:** 2 **Var explained:** 0.625 **Signal:** FLAT

### Signal logic:

- Latest buy\_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell\_cross: 2026-08-05
- Current 252d return: -33.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2025-04-08)



### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Currency            | 1         | 0.5        |
| Leveraged / Inverse | 1         | 0.5        |

**Replication basket** – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

| ticker | rep_wt | name                                  | category            | etf_signal |
|--------|--------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.506  | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.495  | Invesco DB USD Index Bullish Fund ETF | Currency            | FLAT       |

**Members** (PC\_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

| ticker | PC_wt | name                                  | category            | etf_signal |
|--------|-------|---------------------------------------|---------------------|------------|
| SQQQ   | 0.5   | ProShares UltraPro Short QQQ          | Leveraged / Inverse | FLAT       |
| UUP    | 0.5   | Invesco DB USD Index Bullish Fund ETF | Currency            | FLAT       |

**Long ETF Positions (full)**

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

| ticker | cluster    | theme                      | entry      | days | return | p_sell | nss  |
|--------|------------|----------------------------|------------|------|--------|--------|------|
| VHT    | cluster_10 | Healthcare                 | 2026-05-19 | 68   | 19.1%  | 0.00   | 1.00 |
| IYH    | cluster_10 | Healthcare                 | 2026-05-20 | 67   | 18.6%  | 0.00   | 1.00 |
| FHLC   | cluster_10 | Healthcare                 | 2026-05-19 | 68   | 19.0%  | 0.00   | 1.00 |
| PTH    | cluster_10 | Healthcare                 | 2026-06-17 | 48   | 20.5%  | 0.00   | 1.00 |
| XLV    | cluster_10 | Healthcare                 | 2026-05-21 | 66   | 17.1%  | 0.00   | 1.00 |
| PPH    | cluster_10 | Healthcare                 | 2026-05-21 | 66   | 11.6%  | 0.00   | 1.00 |
| NANR   | cluster_11 | Materials                  | 2026-07-24 | 23   | 11.3%  | 0.00   | 1.00 |
| GNR    | cluster_11 | Materials                  | 2026-07-27 | 22   | 9.9%   | 0.00   | 1.00 |
| IGE    | cluster_11 | Materials                  | 2026-07-23 | 24   | 8.9%   | 0.00   | 1.00 |
| GUNR   | cluster_11 | Materials                  | 2026-07-27 | 22   | 8.7%   | 0.00   | 1.00 |
| NORW   | cluster_11 | Materials                  | 2026-08-05 | 15   | 5.5%   | 0.00   | 1.00 |
| MXI    | cluster_11 | Materials                  | 2026-08-05 | 15   | 5.1%   | 0.00   | 1.00 |
| COPX   | cluster_11 | Materials                  | 2026-08-05 | 15   | 10.2%  | 0.00   | 1.00 |
| PICK   | cluster_11 | Materials                  | 2026-08-05 | 15   | 5.0%   | 0.00   | 1.00 |
| URA    | cluster_11 | Materials                  | 2026-08-11 | 11   | 6.4%   | 0.00   | 1.00 |
| XME    | cluster_11 | Materials                  | 2026-08-10 | 12   | 2.4%   | 0.01   | 0.99 |
| DBB    | cluster_11 | Materials                  | 2026-08-04 | 16   | 0.9%   | 0.31   | 0.69 |
| MLPA   | cluster_11 | Materials                  | 2026-07-10 | 33   | 5.0%   | 0.60   | 0.40 |
| AMLP   | cluster_11 | Materials                  | 2026-07-10 | 33   | 5.3%   | 0.85   | 0.15 |
| EMLP   | cluster_11 | Materials                  | 2026-08-13 | 9    | 0.7%   | 0.46   | 0.10 |
| MLPX   | cluster_11 | Materials                  | 2026-08-17 | 7    | 1.0%   | 0.37   | 0.07 |
| GDV    | cluster_17 | Gold                       | 2026-08-06 | 14   | 22.0%  | 0.00   | 1.00 |
| IAU    | cluster_17 | Gold                       | 2026-08-07 | 13   | 5.7%   | 0.00   | 1.00 |
| GLD    | cluster_17 | Gold                       | 2026-08-07 | 13   | 5.7%   | 0.00   | 1.00 |
| SGOL   | cluster_17 | Gold                       | 2026-08-07 | 13   | 5.8%   | 0.00   | 1.00 |
| DBP    | cluster_17 | Gold                       | 2026-08-07 | 13   | 5.9%   | 0.00   | 1.00 |
| GOAU   | cluster_17 | Gold                       | 2026-08-06 | 14   | 21.4%  | 0.00   | 1.00 |
| SIL    | cluster_17 | Gold                       | 2026-08-07 | 13   | 13.7%  | 0.00   | 1.00 |
| GLTR   | cluster_17 | Gold                       | 2026-08-10 | 12   | 4.0%   | 0.00   | 1.00 |
| GDVJ   | cluster_17 | Gold                       | 2026-08-06 | 14   | 21.4%  | 0.00   | 1.00 |
| SLV    | cluster_17 | Gold                       | 2026-08-10 | 12   | 3.7%   | 0.00   | 1.00 |
| SIVR   | cluster_17 | Gold                       | 2026-08-10 | 12   | 3.7%   | 0.00   | 1.00 |
| SILJ   | cluster_17 | Gold                       | 2026-08-07 | 13   | 11.0%  | 0.00   | 1.00 |
| SRLN   | cluster_23 | Bank Loan / Floating Ra... | 2026-07-30 | 19   | 0.8%   | 0.00   | 1.00 |
| IGBH   | cluster_23 | Bank Loan / Floating Ra... | 2026-08-25 | 1    | 0.1%   | 0.00   | 1.00 |
| BKLN   | cluster_23 | Bank Loan / Floating Ra... | 2026-08-05 | 15   | 0.2%   | 0.03   | 0.97 |
| FLTR   | cluster_23 | Bank Loan / Floating Ra... | 2026-08-21 | 3    | -0.0%  | 0.06   | 0.77 |

|      |            |                            |            |     |       |      |      |
|------|------------|----------------------------|------------|-----|-------|------|------|
| FLOT | cluster_23 | Bank Loan / Floating Ra... | 2026-08-21 | 3   | 0.0%  | 0.07 | 0.77 |
| FLRN | cluster_23 | Bank Loan / Floating Ra... | 2026-08-24 | 2   | 0.1%  | 0.12 | 0.56 |
| EZA  | cluster_7  | Emerging Markets Equity    | 2026-08-05 | 15  | 7.2%  | 0.00 | 1.00 |
| DVYE | cluster_7  | Emerging Markets Equity    | 2026-07-24 | 23  | 4.1%  | 0.00 | 1.00 |
| FEMS | cluster_7  | Emerging Markets Equity    | 2026-08-12 | 10  | 2.5%  | 0.00 | 1.00 |
| ECH  | cluster_7  | Emerging Markets Equity    | 2026-08-05 | 15  | 0.8%  | 0.00 | 1.00 |
| FEM  | cluster_7  | Emerging Markets Equity    | 2026-08-07 | 13  | 2.5%  | 0.00 | 1.00 |
| DEM  | cluster_7  | Emerging Markets Equity    | 2026-08-07 | 13  | 1.5%  | 0.01 | 0.99 |
| EEMV | cluster_7  | Emerging Markets Equity    | 2026-08-12 | 10  | 0.9%  | 0.01 | 0.99 |
| ESGE | cluster_7  | Emerging Markets Equity    | 2026-08-14 | 8   | 1.0%  | 0.01 | 0.99 |
| EEM  | cluster_7  | Emerging Markets Equity    | 2026-08-14 | 8   | 0.8%  | 0.01 | 0.99 |
| EMGF | cluster_7  | Emerging Markets Equity    | 2026-08-14 | 8   | 0.9%  | 0.01 | 0.99 |
| IEMG | cluster_7  | Emerging Markets Equity    | 2026-08-13 | 9   | 0.7%  | 0.01 | 0.99 |
| AAXJ | cluster_7  | Emerging Markets Equity    | 2026-08-14 | 8   | 0.8%  | 0.01 | 0.99 |
| EWX  | cluster_7  | Emerging Markets Equity    | 2026-08-14 | 8   | 1.7%  | 0.01 | 0.99 |
| GEM  | cluster_7  | Emerging Markets Equity    | 2026-08-13 | 9   | 0.5%  | 0.01 | 0.98 |
| XSOE | cluster_7  | Emerging Markets Equity    | 2026-08-25 | 1   | -0.1% | 0.02 | 0.97 |
| EEMA | cluster_7  | Emerging Markets Equity    | 2026-08-12 | 10  | 0.7%  | 0.03 | 0.96 |
| SPEM | cluster_7  | Emerging Markets Equity    | 2026-08-06 | 14  | 1.4%  | 0.06 | 0.94 |
| FNDE | cluster_7  | Emerging Markets Equity    | 2026-07-30 | 19  | 2.2%  | 0.06 | 0.94 |
| SCHE | cluster_7  | Emerging Markets Equity    | 2026-08-05 | 15  | 1.1%  | 0.07 | 0.93 |
| VWO  | cluster_7  | Emerging Markets Equity    | 2026-08-06 | 14  | 1.2%  | 0.08 | 0.92 |
| SDIV | cluster_7  | Emerging Markets Equity    | 2026-08-19 | 5   | -0.2% | 0.08 | 0.92 |
| AIA  | cluster_7  | Emerging Markets Equity    | 2026-08-14 | 8   | -0.2% | 0.06 | 0.92 |
| EWT  | cluster_7  | Emerging Markets Equity    | 2026-08-11 | 11  | 2.4%  | 0.08 | 0.88 |
| DGS  | cluster_7  | Emerging Markets Equity    | 2026-08-25 | 1   | -0.2% | 0.08 | 0.86 |
| EWI  | cluster_7  | Emerging Markets Equity    | 2026-08-25 | 1   | -0.4% | 0.14 | 0.85 |
| PXH  | cluster_7  | Emerging Markets Equity    | 2026-07-31 | 18  | 1.5%  | 0.20 | 0.80 |
| GMF  | cluster_7  | Emerging Markets Equity    | 2026-08-06 | 14  | 0.7%  | 0.31 | 0.66 |
| ILF  | cluster_7  | Emerging Markets Equity    | 2026-08-25 | 1   | -0.5% | 0.28 | 0.56 |
| EWZ  | cluster_7  | Emerging Markets Equity    | 2026-08-26 | 0   | 0.0%  | 0.47 | 0.04 |
| SCHD | cluster_8  | U.S. Dividend Equity       | 2026-07-07 | 36  | 7.7%  | 0.00 | 1.00 |
| HDV  | cluster_8  | U.S. Dividend Equity       | 2026-06-25 | 43  | 8.0%  | 0.00 | 1.00 |
| PWV  | cluster_8  | U.S. Dividend Equity       | 2026-04-02 | 100 | 17.9% | 0.00 | 1.00 |
| FDL  | cluster_8  | U.S. Dividend Equity       | 2026-07-16 | 29  | 4.3%  | 0.00 | 1.00 |
| FTCS | cluster_8  | U.S. Dividend Equity       | 2026-06-26 | 42  | 8.3%  | 0.01 | 0.99 |
| FTA  | cluster_8  | U.S. Dividend Equity       | 2026-05-21 | 66  | 10.5% | 0.03 | 0.97 |
| ACWV | cluster_8  | U.S. Dividend Equity       | 2026-07-06 | 37  | 4.3%  | 0.03 | 0.97 |
| SPHD | cluster_8  | U.S. Dividend Equity       | 2026-06-08 | 55  | 6.8%  | 0.03 | 0.97 |
| SDY  | cluster_8  | U.S. Dividend Equity       | 2026-05-20 | 67  | 7.0%  | 0.07 | 0.93 |
| OUSA | cluster_8  | U.S. Dividend Equity       | 2026-07-01 | 39  | 6.3%  | 0.12 | 0.88 |
| FVD  | cluster_8  | U.S. Dividend Equity       | 2026-06-10 | 53  | 6.8%  | 0.34 | 0.66 |
| LGLV | cluster_8  | U.S. Dividend Equity       | 2026-06-26 | 42  | 4.0%  | 0.49 | 0.51 |
| FXB  | cluster_25 | International Fixed Inc... | 2026-07-31 | 18  | 0.8%  | 0.00 | 1.00 |
| FXE  | cluster_25 | International Fixed Inc... | 2026-08-04 | 16  | 1.2%  | 0.00 | 1.00 |

|      |            |                            |            |    |       |      |      |
|------|------------|----------------------------|------------|----|-------|------|------|
| FXF  | cluster_25 | International Fixed Inc... | 2026-08-20 | 4  | -0.6% | 0.00 | 1.00 |
| IGOV | cluster_25 | International Fixed Inc... | 2026-08-24 | 2  | 0.0%  | 0.03 | 0.97 |
| NEAR | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-18 | 6  | 0.1%  | 0.00 | 1.00 |
| FTSM | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-17 | 7  | 0.1%  | 0.02 | 0.95 |
| ICSH | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-17 | 7  | 0.1%  | 0.07 | 0.90 |
| GBIL | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-18 | 6  | 0.1%  | 0.08 | 0.86 |
| MINT | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-17 | 7  | 0.1%  | 0.08 | 0.85 |
| SHV  | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-18 | 6  | 0.1%  | 0.09 | 0.84 |
| VNLA | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-21 | 3  | 0.0%  | 0.04 | 0.82 |
| BIL  | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-18 | 6  | 0.1%  | 0.11 | 0.77 |
| GSY  | cluster_24 | Cash / Ultra-Short Fixe... | 2026-08-07 | 13 | -0.1% | 0.40 | 0.51 |
| USO  | cluster_22 | Broad Commodities          | 2026-08-11 | 11 | -0.2% | 0.22 | 0.78 |
| DBO  | cluster_22 | Broad Commodities          | 2026-08-11 | 11 | 0.1%  | 0.26 | 0.73 |
| FXA  | cluster_2  | Developed ex-US Equity     | 2026-08-04 | 16 | 1.9%  | 0.00 | 1.00 |
| FXC  | cluster_2  | Developed ex-US Equity     | 2026-08-19 | 5  | -0.4% | 0.00 | 1.00 |
| ENZL | cluster_2  | Developed ex-US Equity     | 2026-07-07 | 36 | 6.8%  | 0.00 | 1.00 |
| VIDI | cluster_2  | Developed ex-US Equity     | 2026-08-10 | 12 | 1.9%  | 0.00 | 1.00 |
| EWK  | cluster_2  | Developed ex-US Equity     | 2026-08-18 | 6  | 2.7%  | 0.00 | 1.00 |
| EWC  | cluster_2  | Developed ex-US Equity     | 2026-07-10 | 33 | 6.1%  | 0.00 | 1.00 |
| IQDF | cluster_2  | Developed ex-US Equity     | 2026-08-04 | 16 | 1.7%  | 0.00 | 1.00 |
| EWS  | cluster_2  | Developed ex-US Equity     | 2026-06-17 | 48 | 15.5% | 0.00 | 1.00 |
| SCHC | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 2.9%  | 0.00 | 1.00 |
| FEP  | cluster_2  | Developed ex-US Equity     | 2026-07-31 | 18 | 2.8%  | 0.00 | 1.00 |
| DLS  | cluster_2  | Developed ex-US Equity     | 2026-08-03 | 17 | 3.2%  | 0.01 | 0.99 |
| FNDF | cluster_2  | Developed ex-US Equity     | 2026-07-31 | 18 | 3.0%  | 0.01 | 0.99 |
| IQLT | cluster_2  | Developed ex-US Equity     | 2026-07-01 | 39 | 6.3%  | 0.01 | 0.99 |
| JPIN | cluster_2  | Developed ex-US Equity     | 2026-07-27 | 22 | 4.5%  | 0.01 | 0.99 |
| GWX  | cluster_2  | Developed ex-US Equity     | 2026-08-06 | 14 | 3.3%  | 0.01 | 0.99 |
| NFRA | cluster_2  | Developed ex-US Equity     | 2026-07-16 | 29 | 1.5%  | 0.02 | 0.98 |
| PXF  | cluster_2  | Developed ex-US Equity     | 2026-08-03 | 17 | 3.0%  | 0.02 | 0.98 |
| VSS  | cluster_2  | Developed ex-US Equity     | 2026-08-07 | 13 | 1.1%  | 0.02 | 0.98 |
| TLTD | cluster_2  | Developed ex-US Equity     | 2026-07-30 | 19 | 3.2%  | 0.02 | 0.98 |
| IDV  | cluster_2  | Developed ex-US Equity     | 2026-07-28 | 21 | 3.3%  | 0.03 | 0.97 |
| SPDW | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.6%  | 0.03 | 0.97 |
| HDEF | cluster_2  | Developed ex-US Equity     | 2026-07-09 | 34 | 5.1%  | 0.04 | 0.96 |
| SCHF | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.4%  | 0.04 | 0.96 |
| VXUS | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.5%  | 0.04 | 0.96 |
| VEA  | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.5%  | 0.04 | 0.96 |
| IXUS | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.4%  | 0.04 | 0.96 |
| QEFA | cluster_2  | Developed ex-US Equity     | 2026-07-06 | 37 | 4.2%  | 0.04 | 0.96 |
| VEU  | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.4%  | 0.04 | 0.96 |
| ACWX | cluster_2  | Developed ex-US Equity     | 2026-08-05 | 15 | 1.3%  | 0.04 | 0.96 |
| EFAV | cluster_2  | Developed ex-US Equity     | 2026-07-16 | 29 | 4.5%  | 0.04 | 0.96 |
| EWU  | cluster_2  | Developed ex-US Equity     | 2026-07-16 | 29 | 4.0%  | 0.05 | 0.95 |
| FNDC | cluster_2  | Developed ex-US Equity     | 2026-08-04 | 16 | 1.9%  | 0.07 | 0.93 |

|      |            |            |        |        |            |    |       |      |       |
|------|------------|------------|--------|--------|------------|----|-------|------|-------|
| VYMI | cluster_2  | Developed  | ex-US  | Equity | 2026-07-08 | 35 | 6.1%  | 0.08 | 0.92  |
| CWI  | cluster_2  | Developed  | ex-US  | Equity | 2026-08-04 | 16 | 1.1%  | 0.08 | 0.92  |
| SCZ  | cluster_2  | Developed  | ex-US  | Equity | 2026-08-04 | 16 | 2.0%  | 0.08 | 0.92  |
| EWL  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-28 | 21 | 1.5%  | 0.08 | 0.91  |
| EPP  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-15 | 30 | 4.6%  | 0.10 | 0.90  |
| VGK  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-01 | 39 | 5.6%  | 0.10 | 0.90  |
| INTF | cluster_2  | Developed  | ex-US  | Equity | 2026-07-27 | 22 | 4.2%  | 0.11 | 0.89  |
| IMTM | cluster_2  | Developed  | ex-US  | Equity | 2026-08-05 | 15 | 1.2%  | 0.12 | 0.88  |
| IEV  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-27 | 22 | 4.1%  | 0.12 | 0.88  |
| EWG  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-31 | 18 | 3.4%  | 0.12 | 0.88  |
| IEUR | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 1.9%  | 0.13 | 0.87  |
| EWD  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 2.8%  | 0.13 | 0.87  |
| DWX  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-09 | 34 | 3.7%  | 0.13 | 0.87  |
| HFXI | cluster_2  | Developed  | ex-US  | Equity | 2026-08-05 | 15 | 0.9%  | 0.13 | 0.87  |
| IVLU | cluster_2  | Developed  | ex-US  | Equity | 2026-07-27 | 22 | 3.8%  | 0.14 | 0.86  |
| EPOL | cluster_2  | Developed  | ex-US  | Equity | 2026-07-08 | 35 | 12.4% | 0.15 | 0.85  |
| BBAX | cluster_2  | Developed  | ex-US  | Equity | 2026-07-14 | 31 | 5.2%  | 0.15 | 0.85  |
| RODM | cluster_2  | Developed  | ex-US  | Equity | 2026-07-27 | 22 | 2.4%  | 0.16 | 0.84  |
| GSIE | cluster_2  | Developed  | ex-US  | Equity | 2026-07-02 | 38 | 5.0%  | 0.17 | 0.83  |
| EFV  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-08 | 35 | 5.9%  | 0.18 | 0.82  |
| IEFA | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 2.1%  | 0.18 | 0.82  |
| DWM  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 1.9%  | 0.20 | 0.80  |
| EFA  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 1.9%  | 0.21 | 0.79  |
| EFG  | cluster_2  | Developed  | ex-US  | Equity | 2026-08-05 | 15 | 0.7%  | 0.25 | 0.75  |
| ESGD | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 1.8%  | 0.26 | 0.74  |
| EWA  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-20 | 27 | 5.7%  | 0.26 | 0.74  |
| EZU  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-31 | 18 | 2.6%  | 0.27 | 0.73  |
| EWO  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-31 | 18 | 4.4%  | 0.29 | 0.71  |
| FEZ  | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 2.8%  | 0.30 | 0.70  |
| EUFN | cluster_2  | Developed  | ex-US  | Equity | 2026-06-12 | 51 | 9.7%  | 0.43 | 0.57  |
| EWN  | cluster_2  | Developed  | ex-US  | Equity | 2026-08-05 | 15 | 0.3%  | 0.39 | 0.56  |
| IDLV | cluster_2  | Developed  | ex-US  | Equity | 2026-07-10 | 33 | 3.3%  | 0.53 | 0.47  |
| VIGI | cluster_2  | Developed  | ex-US  | Equity | 2026-06-30 | 40 | 5.5%  | 0.64 | 0.36  |
| VNQI | cluster_2  | Developed  | ex-US  | Equity | 2026-08-24 | 2  | -0.3% | 0.63 | 0.20  |
| HEZU | cluster_2  | Developed  | ex-US  | Equity | 2026-07-30 | 19 | 1.6%  | 0.72 | 0.19  |
| IPKW | cluster_2  | Developed  | ex-US  | Equity | 2026-07-10 | 33 | 4.3%  | 0.84 | 0.16  |
| ARKK | cluster_15 | Technology |        |        | 2026-08-10 | 12 | 6.6%  | 0.00 | 1.00  |
| ARKW | cluster_15 | Technology |        |        | 2026-08-21 | 3  | -0.3% | 0.00 | 1.00  |
| BLOK | cluster_15 | Technology |        |        | 2026-08-25 | 1  | -1.6% | 0.00 | 1.00  |
| SKYY | cluster_15 | Technology |        |        | 2026-07-30 | 19 | 14.2% | 0.20 | 0.80  |
| FDN  | cluster_15 | Technology |        |        | 2026-08-03 | 17 | 1.9%  | 0.34 | 0.66  |
| ARKQ | cluster_15 | Technology |        |        | 2026-08-11 | 11 | -4.7% | 0.50 | 0.33  |
| IPO  | cluster_15 | Technology |        |        | 2026-08-13 | 9  | -5.1% | 0.48 | -0.21 |
| TUR  | cluster_18 | India      | Equity |        | 2026-08-14 | 8  | 2.5%  | 0.00 | 1.00  |
| EWM  | cluster_18 | India      | Equity |        | 2026-07-30 | 19 | 2.6%  | 0.00 | 1.00  |

|      |            |                      |            |    |       |      |       |
|------|------------|----------------------|------------|----|-------|------|-------|
| VNM  | cluster_18 | India Equity         | 2026-08-26 | 0  | 0.0%  | 0.02 | 0.97  |
| IYM  | cluster_4  | U.S. Large Cap Value | 2026-08-06 | 14 | 6.3%  | 0.00 | 1.00  |
| VAW  | cluster_4  | U.S. Large Cap Value | 2026-08-06 | 14 | 3.2%  | 0.01 | 0.99  |
| RSPM | cluster_4  | U.S. Large Cap Value | 2026-07-27 | 22 | 4.5%  | 0.02 | 0.98  |
| XLB  | cluster_4  | U.S. Large Cap Value | 2026-07-28 | 21 | 2.5%  | 0.02 | 0.98  |
| VLUE | cluster_4  | U.S. Large Cap Value | 2026-08-07 | 13 | 2.1%  | 0.03 | 0.97  |
| VTV  | cluster_4  | U.S. Large Cap Value | 2026-04-13 | 94 | 12.0% | 0.05 | 0.95  |
| RSP  | cluster_4  | U.S. Large Cap Value | 2026-04-14 | 93 | 11.2% | 0.07 | 0.93  |
| FNDX | cluster_4  | U.S. Large Cap Value | 2026-04-09 | 96 | 13.2% | 0.12 | 0.88  |
| PKW  | cluster_4  | U.S. Large Cap Value | 2026-05-20 | 67 | 11.2% | 0.13 | 0.87  |
| PRF  | cluster_4  | U.S. Large Cap Value | 2026-04-09 | 96 | 13.7% | 0.20 | 0.80  |
| IWD  | cluster_4  | U.S. Large Cap Value | 2026-04-10 | 95 | 16.5% | 0.26 | 0.74  |
| VONV | cluster_4  | U.S. Large Cap Value | 2026-04-10 | 95 | 16.5% | 0.27 | 0.73  |
| DGRO | cluster_4  | U.S. Large Cap Value | 2026-04-13 | 94 | 10.3% | 0.29 | 0.71  |
| IWX  | cluster_4  | U.S. Large Cap Value | 2026-04-10 | 95 | 17.6% | 0.30 | 0.70  |
| DLN  | cluster_4  | U.S. Large Cap Value | 2026-04-13 | 94 | 9.8%  | 0.33 | 0.67  |
| DTD  | cluster_4  | U.S. Large Cap Value | 2026-04-13 | 94 | 9.6%  | 0.35 | 0.65  |
| SCHV | cluster_4  | U.S. Large Cap Value | 2026-08-04 | 16 | 0.1%  | 0.42 | 0.56  |
| JVAL | cluster_4  | U.S. Large Cap Value | 2026-07-30 | 19 | 3.4%  | 0.44 | 0.56  |
| IYZ  | cluster_4  | U.S. Large Cap Value | 2026-08-06 | 14 | 0.8%  | 0.47 | 0.53  |
| RWL  | cluster_4  | U.S. Large Cap Value | 2026-04-10 | 95 | 13.7% | 0.55 | 0.45  |
| FEX  | cluster_4  | U.S. Large Cap Value | 2026-04-09 | 96 | 10.4% | 0.57 | 0.40  |
| VIG  | cluster_4  | U.S. Large Cap Value | 2026-04-14 | 93 | 9.0%  | 0.65 | 0.35  |
| CGW  | cluster_4  | U.S. Large Cap Value | 2026-08-25 | 1  | 0.2%  | 0.71 | -0.23 |
| ITB  | cluster_4  | U.S. Large Cap Value | 2026-07-28 | 21 | -3.0% | 0.72 | -0.64 |
| AIVL | cluster_4  | U.S. Large Cap Value | 2026-04-10 | 95 | 12.2% |      |       |
| VPL  | cluster_21 | Japan Equity         | 2026-08-10 | 12 | 2.3%  | 0.07 | 0.93  |
| IPAC | cluster_21 | Japan Equity         | 2026-08-03 | 17 | 2.7%  | 0.43 | 0.57  |
| BBJP | cluster_21 | Japan Equity         | 2026-08-05 | 15 | 0.3%  | 0.54 | 0.41  |
| EWJ  | cluster_21 | Japan Equity         | 2026-08-05 | 15 | 0.3%  | 0.55 | 0.39  |
| FLJH | cluster_21 | Japan Equity         | 2026-08-26 | 0  | 0.0%  | 0.44 | 0.21  |
| EWJV | cluster_21 | Japan Equity         | 2026-07-06 | 37 | 2.7%  | 0.79 | 0.21  |
| DBJP | cluster_21 | Japan Equity         | 2026-08-10 | 12 | -0.5% | 0.47 | 0.21  |
| HEWJ | cluster_21 | Japan Equity         | 2026-08-26 | 0  | 0.0%  | 0.51 | 0.11  |
| DXJ  | cluster_21 | Japan Equity         | 2026-08-05 | 15 | 1.2%  | 0.58 | 0.09  |
| EMLC | cluster_9  | High Yield Bond      | 2026-08-05 | 15 | 0.9%  | 0.00 | 1.00  |
| SHYG | cluster_9  | High Yield Bond      | 2026-08-25 | 1  | -0.0% | 0.00 | 1.00  |
| EBND | cluster_9  | High Yield Bond      | 2026-07-31 | 18 | 1.5%  | 0.00 | 1.00  |
| JNK  | cluster_9  | High Yield Bond      | 2026-08-25 | 1  | -0.0% | 0.00 | 1.00  |
| HYG  | cluster_9  | High Yield Bond      | 2026-08-25 | 1  | -0.0% | 0.00 | 1.00  |
| BBHY | cluster_9  | High Yield Bond      | 2026-08-26 | 0  | 0.0%  | 0.00 | 1.00  |
| HYEM | cluster_9  | High Yield Bond      | 2026-08-25 | 1  | -0.1% | 0.00 | 1.00  |
| HYLB | cluster_9  | High Yield Bond      | 2026-08-25 | 1  | -0.0% | 0.00 | 0.99  |
| ANGL | cluster_9  | High Yield Bond      | 2026-08-26 | 0  | 0.0%  | 0.00 | 0.97  |
| PFXF | cluster_9  | High Yield Bond      | 2026-08-10 | 12 | 0.2%  | 0.44 | 0.55  |

|      |           |                       |            |    |       |      |      |
|------|-----------|-----------------------|------------|----|-------|------|------|
| PFF  | cluster_9 | High Yield Bond       | 2026-08-13 | 9  | -0.6% | 0.57 | 0.31 |
| FV   | cluster_1 | U.S. Large Cap Equity | 2026-08-10 | 12 | 1.5%  | 0.00 | 1.00 |
| XLC  | cluster_1 | U.S. Large Cap Equity | 2026-08-06 | 14 | 1.3%  | 0.03 | 0.97 |
| QQEW | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | 1.2%  | 0.09 | 0.91 |
| FCOM | cluster_1 | U.S. Large Cap Equity | 2026-08-25 | 1  | -0.6% | 0.09 | 0.86 |
| VOX  | cluster_1 | U.S. Large Cap Equity | 2026-08-25 | 1  | -0.6% | 0.09 | 0.85 |
| AOK  | cluster_1 | U.S. Large Cap Equity | 2026-08-06 | 14 | 0.4%  | 0.15 | 0.84 |
| QAI  | cluster_1 | U.S. Large Cap Equity | 2026-08-06 | 14 | 0.5%  | 0.20 | 0.79 |
| AOM  | cluster_1 | U.S. Large Cap Equity | 2026-08-06 | 14 | 0.3%  | 0.22 | 0.78 |
| XT   | cluster_1 | U.S. Large Cap Equity | 2026-08-07 | 13 | -0.8% | 0.21 | 0.78 |
| EIS  | cluster_1 | U.S. Large Cap Equity | 2026-08-12 | 10 | 0.5%  | 0.23 | 0.77 |
| AOR  | cluster_1 | U.S. Large Cap Equity | 2026-08-05 | 15 | 0.2%  | 0.24 | 0.75 |
| VT   | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | 0.3%  | 0.26 | 0.74 |
| SPGM | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | 0.2%  | 0.26 | 0.74 |
| CRBN | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | 0.1%  | 0.27 | 0.73 |
| ACWI | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | 0.2%  | 0.28 | 0.72 |
| AOA  | cluster_1 | U.S. Large Cap Equity | 2026-08-05 | 15 | 0.1%  | 0.29 | 0.71 |
| DSI  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.5% | 0.30 | 0.70 |
| XLY  | cluster_1 | U.S. Large Cap Equity | 2026-08-05 | 15 | -1.2% | 0.29 | 0.69 |
| IYC  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.5% | 0.32 | 0.67 |
| SUSA | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.2% | 0.38 | 0.61 |
| URTH | cluster_1 | U.S. Large Cap Equity | 2026-08-03 | 17 | 1.5%  | 0.39 | 0.60 |
| DGRW | cluster_1 | U.S. Large Cap Equity | 2026-08-03 | 17 | 1.5%  | 0.40 | 0.60 |
| FDIS | cluster_1 | U.S. Large Cap Equity | 2026-08-05 | 15 | -1.7% | 0.36 | 0.59 |
| VCR  | cluster_1 | U.S. Large Cap Equity | 2026-08-05 | 15 | -1.8% | 0.36 | 0.58 |
| VONE | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.6% | 0.47 | 0.52 |
| IWB  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.6% | 0.47 | 0.52 |
| SPHB | cluster_1 | U.S. Large Cap Equity | 2026-08-26 | 0  | 0.0%  | 0.27 | 0.52 |
| IWV  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.6% | 0.48 | 0.52 |
| VTHR | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.6% | 0.48 | 0.52 |
| EPS  | cluster_1 | U.S. Large Cap Equity | 2026-08-03 | 17 | 1.0%  | 0.48 | 0.51 |
| ILCB | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.6% | 0.49 | 0.50 |
| VV   | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.50 | 0.50 |
| IWL  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.8% | 0.50 | 0.49 |
| IYY  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.50 | 0.49 |
| IVV  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.51 | 0.48 |
| MGC  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.9% | 0.51 | 0.48 |
| SPY  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.51 | 0.48 |
| VTI  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.51 | 0.48 |
| VOO  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.51 | 0.48 |
| ITOT | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.51 | 0.48 |
| SCHB | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.6% | 0.52 | 0.47 |
| SCHX | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.52 | 0.47 |
| ESGU | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -0.7% | 0.52 | 0.47 |
| OEF  | cluster_1 | U.S. Large Cap Equity | 2026-08-04 | 16 | -1.1% | 0.54 | 0.45 |

|      |            |                            |            |    |       |      |       |
|------|------------|----------------------------|------------|----|-------|------|-------|
| SPTM | cluster_1  | U.S. Large Cap Equity      | 2026-08-04 | 16 | -0.7% | 0.55 | 0.44  |
| LRGF | cluster_1  | U.S. Large Cap Equity      | 2026-08-03 | 17 | 1.5%  | 0.55 | 0.44  |
| CATH | cluster_1  | U.S. Large Cap Equity      | 2026-08-04 | 16 | -0.9% | 0.55 | 0.44  |
| GSLC | cluster_1  | U.S. Large Cap Equity      | 2026-08-03 | 17 | 1.2%  | 0.57 | 0.43  |
| JHML | cluster_1  | U.S. Large Cap Equity      | 2026-08-03 | 17 | 1.2%  | 0.57 | 0.42  |
| SPYX | cluster_1  | U.S. Large Cap Equity      | 2026-08-04 | 16 | -0.9% | 0.57 | 0.42  |
| IWP  | cluster_1  | U.S. Large Cap Equity      | 2026-08-07 | 13 | -1.1% | 0.50 | 0.35  |
| VOT  | cluster_1  | U.S. Large Cap Equity      | 2026-08-05 | 15 | -0.1% | 0.57 | 0.29  |
| IMCG | cluster_1  | U.S. Large Cap Equity      | 2026-08-04 | 16 | -0.2% | 0.63 | 0.18  |
| CWB  | cluster_1  | U.S. Large Cap Equity      | 2026-08-14 | 8  | -2.7% | 0.56 | -0.13 |
| SPHQ | cluster_1  | U.S. Large Cap Equity      | 2026-08-14 | 8  | -1.2% | 0.42 | -0.14 |
| ICVT | cluster_1  | U.S. Large Cap Equity      | 2026-08-17 | 7  | -3.0% | 0.49 | -0.15 |
| VFMO | cluster_1  | U.S. Large Cap Equity      | 2026-08-13 | 9  | -2.3% | 0.49 | -0.17 |
| RPG  | cluster_1  | U.S. Large Cap Equity      | 2026-08-17 | 7  | -5.0% | 0.51 | -0.27 |
| KCE  | cluster_13 | Financials                 | 2026-07-07 | 36 | 9.0%  | 0.00 | 1.00  |
| XLF  | cluster_13 | Financials                 | 2026-04-13 | 94 | 12.8% | 0.50 | 0.50  |
| FNCL | cluster_13 | Financials                 | 2026-04-13 | 94 | 12.5% | 0.58 | 0.42  |
| VTIP | cluster_20 | Inflation-Protected Fix... | 2026-08-25 | 1  | -0.0% | 0.00 | 1.00  |
| VGSH | cluster_5  | Aggregate Fixed Income     | 2026-08-17 | 7  | 0.0%  | 0.01 | 0.98  |
| SPSB | cluster_5  | Aggregate Fixed Income     | 2026-08-18 | 6  | 0.1%  | 0.01 | 0.98  |
| SPTS | cluster_5  | Aggregate Fixed Income     | 2026-08-17 | 7  | 0.0%  | 0.02 | 0.97  |
| SHY  | cluster_5  | Aggregate Fixed Income     | 2026-08-17 | 7  | 0.1%  | 0.03 | 0.97  |
| LDUR | cluster_5  | Aggregate Fixed Income     | 2026-08-17 | 7  | 0.1%  | 0.04 | 0.95  |
| SCHO | cluster_5  | Aggregate Fixed Income     | 2026-08-17 | 7  | 0.0%  | 0.05 | 0.92  |
| SLQD | cluster_5  | Aggregate Fixed Income     | 2026-08-25 | 1  | -0.1% | 0.01 | 0.91  |
| BSV  | cluster_5  | Aggregate Fixed Income     | 2026-08-26 | 0  | 0.0%  | 0.02 | 0.88  |
| ISTB | cluster_5  | Aggregate Fixed Income     | 2026-08-26 | 0  | 0.0%  | 0.03 | 0.86  |
| CMBS | cluster_5  | Aggregate Fixed Income     | 2026-08-26 | 0  | 0.0%  | 0.00 | 0.86  |
| IBDR | cluster_5  | Aggregate Fixed Income     | 2026-08-20 | 4  | 0.1%  | 0.10 | 0.78  |
| BSCQ | cluster_5  | Aggregate Fixed Income     | 2026-08-07 | 13 | -0.1% | 0.66 | 0.04  |
| LMBS | cluster_5  | Aggregate Fixed Income     | 2026-08-19 | 5  | -0.3% | 0.38 | -0.01 |
| QYLD | cluster_6  | U.S. Large Cap Growth      | 2026-08-07 | 13 | 0.1%  | 0.07 | 0.93  |
| AIQ  | cluster_6  | U.S. Large Cap Growth      | 2026-08-10 | 12 | -0.7% | 0.17 | 0.81  |
| SCHG | cluster_6  | U.S. Large Cap Growth      | 2026-08-04 | 16 | -0.6% | 0.42 | 0.58  |
| NXTG | cluster_6  | U.S. Large Cap Growth      | 2026-08-12 | 10 | -1.7% | 0.41 | 0.52  |
| FXL  | cluster_6  | U.S. Large Cap Growth      | 2026-08-06 | 14 | 0.9%  | 0.47 | 0.50  |
| PTLC | cluster_6  | U.S. Large Cap Growth      | 2026-08-04 | 16 | -0.7% | 0.52 | 0.47  |
| IXN  | cluster_6  | U.S. Large Cap Growth      | 2026-08-07 | 13 | -1.5% | 0.46 | 0.34  |
| XNTK | cluster_6  | U.S. Large Cap Growth      | 2026-08-13 | 9  | -3.3% | 0.33 | 0.32  |
| QQQ  | cluster_6  | U.S. Large Cap Growth      | 2026-08-07 | 13 | -1.6% | 0.46 | 0.30  |
| PTNQ | cluster_6  | U.S. Large Cap Growth      | 2026-08-07 | 13 | -1.7% | 0.46 | 0.30  |
| QTEC | cluster_6  | U.S. Large Cap Growth      | 2026-08-10 | 12 | -1.0% | 0.43 | 0.26  |
| KURE | cluster_12 | China Equity               | 2026-07-06 | 37 | 11.4% | 0.00 | 1.00  |
| SMMV | cluster_3  | U.S. Mid Cap Equity        | 2026-06-25 | 43 | 5.9%  | 0.04 | 0.96  |
| IWC  | cluster_3  | U.S. Mid Cap Equity        | 2026-08-05 | 15 | 0.2%  | 0.21 | 0.77  |

|      |            |                     |            |    |       |      |       |
|------|------------|---------------------|------------|----|-------|------|-------|
| REM  | cluster_3  | U.S. Mid Cap Equity | 2026-08-12 | 10 | -1.2% | 0.17 | 0.62  |
| SCHM | cluster_3  | U.S. Mid Cap Equity | 2026-08-07 | 13 | -0.2% | 0.33 | 0.57  |
| VBK  | cluster_3  | U.S. Mid Cap Equity | 2026-08-07 | 13 | -1.2% | 0.35 | 0.56  |
| SMLF | cluster_3  | U.S. Mid Cap Equity | 2026-08-03 | 17 | 1.5%  | 0.53 | 0.43  |
| ISCG | cluster_3  | U.S. Mid Cap Equity | 2026-08-04 | 16 | -0.6% | 0.51 | 0.42  |
| VXF  | cluster_3  | U.S. Mid Cap Equity | 2026-08-05 | 15 | 0.2%  | 0.54 | 0.36  |
| IWO  | cluster_3  | U.S. Mid Cap Equity | 2026-08-06 | 14 | 0.2%  | 0.47 | 0.29  |
| VTWG | cluster_3  | U.S. Mid Cap Equity | 2026-08-06 | 14 | 0.2%  | 0.47 | 0.28  |
| VTWO | cluster_3  | U.S. Mid Cap Equity | 2026-08-04 | 16 | -0.9% | 0.56 | 0.28  |
| IWM  | cluster_3  | U.S. Mid Cap Equity | 2026-08-04 | 16 | -0.9% | 0.57 | 0.27  |
| SCHA | cluster_3  | U.S. Mid Cap Equity | 2026-08-07 | 13 | -0.3% | 0.45 | 0.15  |
| IJS  | cluster_3  | U.S. Mid Cap Equity | 2026-04-10 | 95 | 12.2% | 0.85 | 0.05  |
| IYT  | cluster_3  | U.S. Mid Cap Equity | 2026-08-06 | 14 | 1.1%  | 0.27 | 0.02  |
| PBJ  | cluster_19 | Consumer Staples    | 2026-08-18 | 6  | 1.0%  | 0.01 | 0.99  |
| PSCC | cluster_19 | Consumer Staples    | 2026-06-10 | 53 | 10.0% | 0.07 | 0.93  |
| XLP  | cluster_19 | Consumer Staples    | 2026-07-27 | 22 | 1.1%  | 0.08 | 0.92  |
| FSTA | cluster_19 | Consumer Staples    | 2026-07-28 | 21 | -1.3% | 0.24 | 0.75  |
| VDC  | cluster_19 | Consumer Staples    | 2026-07-28 | 21 | -1.1% | 0.26 | 0.73  |
| XLRE | cluster_14 | Real Estate         | 2026-08-18 | 6  | 1.0%  | 0.39 | 0.28  |
| REZ  | cluster_14 | Real Estate         | 2026-08-24 | 2  | -0.6% | 0.28 | 0.07  |
| VNQ  | cluster_14 | Real Estate         | 2026-08-19 | 5  | 0.0%  | 0.44 | 0.02  |
| FREL | cluster_14 | Real Estate         | 2026-08-19 | 5  | -0.1% | 0.47 | -0.02 |
| ICF  | cluster_14 | Real Estate         | 2026-08-25 | 1  | -0.6% | 0.51 | -0.19 |
| SCHH | cluster_14 | Real Estate         | 2026-08-25 | 1  | -0.5% | 0.58 | -0.45 |
| RWR  | cluster_14 | Real Estate         | 2026-08-25 | 1  | -0.6% | 0.59 | -0.52 |
| RWO  | cluster_14 | Real Estate         | 2026-08-25 | 1  | -0.4% | 0.71 | -0.58 |
| REET | cluster_14 | Real Estate         | 2026-08-25 | 1  | -0.6% | 0.77 | -0.66 |
| SUB  | cluster_16 | Municipal Bond      | 2026-08-13 | 9  | 0.0%  | 0.01 | 0.99  |

**ETF Net Signal Strength**

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

**Top 20 most bullish (cluster-deduped, abs-mom approved)**

| ticker | cluster    | theme                        | p_buy | nss  |
|--------|------------|------------------------------|-------|------|
| FXA    | cluster_2  | Developed ex-US Equity       | 1.00  | 1.00 |
| NANR   | cluster_11 | Materials                    | 1.00  | 1.00 |
| ARKG   | cluster_10 | Healthcare                   | 1.00  | 1.00 |
| GDX    | cluster_17 | Gold                         | 1.00  | 1.00 |
| SCHD   | cluster_8  | U.S. Dividend Equity         | 1.00  | 1.00 |
| ARKK   | cluster_15 | Technology                   | 1.00  | 1.00 |
| IAI    | cluster_13 | Financials                   | 1.00  | 1.00 |
| SRLN   | cluster_23 | Bank Loan / Floating Rate    | 1.00  | 1.00 |
| EMLC   | cluster_9  | High Yield Bond              | 1.00  | 1.00 |
| IYM    | cluster_4  | U.S. Large Cap Value         | 1.00  | 1.00 |
| EZA    | cluster_7  | Emerging Markets Equity      | 1.00  | 1.00 |
| FTGC   | cluster_22 | Broad Commodities            | 1.00  | 1.00 |
| NEAR   | cluster_24 | Cash / Ultra-Short Fixed ... | 1.00  | 1.00 |
| FV     | cluster_1  | U.S. Large Cap Equity        | 1.00  | 1.00 |
| SUB    | cluster_16 | Municipal Bond               | 1.00  | 0.99 |
| PBJ    | cluster_19 | Consumer Staples             | 0.99  | 0.99 |
| VGSH   | cluster_5  | Aggregate Fixed Income       | 1.00  | 0.98 |
| SMMV   | cluster_3  | U.S. Mid Cap Equity          | 1.00  | 0.96 |
| VPL    | cluster_21 | Japan Equity                 | 1.00  | 0.93 |
| QYLD   | cluster_6  | U.S. Large Cap Growth        | 0.99  | 0.93 |

**Top 20 most bearish (cluster-deduped)**

| ticker | cluster    | theme                  | p_sell | nss   |
|--------|------------|------------------------|--------|-------|
| IAT    | cluster_13 | Financials             | 1.00   | -1.00 |
| SMDV   | cluster_3  | U.S. Mid Cap Equity    | 1.00   | -1.00 |
| PRN    | cluster_1  | U.S. Large Cap Equity  | 0.99   | -0.99 |
| UUP    | cluster_26 | Leveraged / Inverse    | 0.99   | -0.99 |
| PAVE   | cluster_4  | U.S. Large Cap Value   | 1.00   | -0.99 |
| IGF    | cluster_2  | Developed ex-US Equity | 0.99   | -0.99 |

|      |            |                              |      |       |
|------|------------|------------------------------|------|-------|
| SPLV | cluster_8  | U.S. Dividend Equity         | 0.99 | -0.99 |
| EPHE | cluster_18 | India Equity                 | 1.00 | -0.99 |
| FPE  | cluster_9  | High Yield Bond              | 0.98 | -0.98 |
| CQQQ | cluster_12 | China Equity                 | 0.98 | -0.98 |
| XLU  | cluster_19 | Consumer Staples             | 0.96 | -0.96 |
| PSCT | cluster_6  | U.S. Large Cap Growth        | 0.96 | -0.96 |
| INDS | cluster_14 | Real Estate                  | 0.96 | -0.93 |
| PZA  | cluster_16 | Municipal Bond               | 0.85 | -0.85 |
| BNDX | cluster_5  | Aggregate Fixed Income       | 0.79 | -0.79 |
| IGHG | cluster_23 | Bank Loan / Floating Rate    | 0.64 | -0.59 |
| HACK | cluster_15 | Technology                   | 0.89 | -0.26 |
| TDTT | cluster_20 | Inflation-Protected Fixed... | 0.13 | -0.02 |
| EWZ  | cluster_7  | Emerging Markets Equity      | 0.47 | 0.04  |
| MLPX | cluster_11 | Materials                    | 0.37 | 0.07  |

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## Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

| ticker | name                               | category                 | cluster_id | cluster_size | etf_signal | max_corr_to_others | median_corr_to_others | abs_pc1_loading | compo |
|--------|------------------------------------|--------------------------|------------|--------------|------------|--------------------|-----------------------|-----------------|-------|
| UNG    | United States Natural Gas Fund ... | Natural Gas              | cluster_18 | 12           | FLAT       | 0.487              | 0.071                 | 0.00016         |       |
| PPLT   | abrdn Physical Platinum Shares ... | Platinum                 | cluster_18 | 12           | FLAT       | 0.302              | 0.094                 | 0.00021         |       |
| FXV    | Invesco CurrencyShares Japanese... | Currency                 | cluster_25 | 6            | FLAT       | 0.730              | 0.096                 | 0.00024         |       |
| VNLA   | Janus Henderson Short Duration ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.647              | 0.206                 | 0.00036         |       |
| GBIL   | Goldman Sachs Access Treasury 0... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.759              | 0.029                 | 0.00007         |       |
| DBA    | Invesco DB Agriculture Fund        | Agricultural Commodit... | cluster_18 | 12           | FLAT       | 0.591              | 0.168                 | 0.00039         |       |
| FTSM   | First Trust Enhanced Short Matu... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.734              | 0.120                 | 0.00031         |       |
| ICSH   | iShares Ultra Short Duration Bo... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.734              | 0.142                 | 0.00039         |       |
| GSY    | Invesco Ultra Short Duration ETF   | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.715              | 0.181                 | 0.00044         |       |
| MINT   | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.694              | 0.167                 | 0.00045         |       |
| NEAR   | iShares Short Duration Bond Act... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.699              | 0.187                 | 0.00046         |       |
| TUR    | iShares MSCI Turkey ETF            | Turkey Equity            | cluster_18 | 12           | LONG       | 0.510              | 0.216                 | 0.00055         |       |
| LDUR   | PIMCO Enhanced Low Duration Act... | Fixed Income             | cluster_5  | 58           | LONG       | 0.847              | 0.125                 | 0.00037         |       |
| BIL    | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.882              | 0.025                 | 0.00001         |       |
| SHV    | iShares 0-1 Year Treasury Bond ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.882              | 0.055                 | 0.00018         |       |
| CMBS   | iShares CMBS Bond ETF              | Mortgage-Backed Fixed... | cluster_5  | 58           | LONG       | 0.877              | 0.167                 | 0.00032         |       |
| FXF    | Invesco CurrencyShares Swiss Fr... | Currency                 | cluster_25 | 6            | LONG       | 0.820              | 0.202                 | 0.00049         |       |
| LMBS   | First Trust Low Duration Opport... | Mortgage-Backed Fixed... | cluster_5  | 58           | LONG       | 0.833              | 0.239                 | 0.00049         |       |
| NOBL   | ProShares S&P 500 Dividend Aris... | U.S. Dividend Equity     | cluster_18 | 12           | FLAT       | 0.447              | 0.270                 | 0.00074         |       |
| SHM    | State Street SPDR Nuveen ICE Sh... | Municipal Bond           | cluster_16 | 13           | FLAT       | 0.878              | 0.250                 | 0.00047         |       |
| VNM    | VanEck Vietnam ETF                 | Vietnam Equity           | cluster_18 | 12           | LONG       | 0.444              | 0.328                 | 0.00083         |       |
| KURE   | KraneShares MSCI All China Heal... | China Equity             | cluster_12 | 15           | LONG       | 0.781              | 0.276                 | 0.00073         |       |
| SUB    | iShares Short-Term National Mun... | Municipal Bond           | cluster_16 | 13           | LONG       | 0.878              | 0.265                 | 0.00055         |       |
| EIDO   | iShares MSCI Indonesia ETF         | Indonesia Equity         | cluster_18 | 12           | FLAT       | 0.500              | 0.350                 | 0.00101         |       |
| DBB    | Invesco DB Base Metals Fund        | Industrial Metals        | cluster_11 | 17           | LONG       | 0.742              | 0.352                 | 0.00084         |       |
| KBA    | KraneShares Bosera MSCI China A... | China Equity             | cluster_12 | 15           | FLAT       | 0.853              | 0.255                 | 0.00073         |       |
| FLTR   | VanEck IG Floating Rate ETF        | Floating Rate Fixed I... | cluster_23 | 9            | LONG       | 0.867              | 0.323                 | 0.00071         |       |
| EPHE   | iShares MSCI Philippines ETF       | Philippines Equity       | cluster_18 | 12           | FLAT       | 0.560              | 0.408                 | 0.00108         |       |
| CNXT   | VanEck ChiNext Innovators ETF      | China Equity             | cluster_12 | 15           | FLAT       | 0.903              | 0.237                 | 0.00064         |       |
| LTPZ   | Pimco 15 Year U.S. TIPS Index E... | Inflation-Protected F... | cluster_20 | 10           | FLAT       | 0.918              | 0.233                 | 0.00059         |       |

## Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

|                          | scope  | parameter                | value |
|--------------------------|--------|--------------------------|-------|
| short_ma                 | screen | short_ma                 | 5     |
| buy_ma                   | screen | buy_ma                   | 50    |
| sell_ma                  | screen | sell_ma                  | 20    |
| vol_window               | screen | vol_window               | 20    |
| horizon                  | screen | horizon                  | 5     |
| imminent_horizon         | screen | imminent_horizon         | 1     |
| cluster_smoothing_window | screen | cluster_smoothing_window | 3     |
| sell_cushion_k           | screen | sell_cushion_k           | 0.5   |
| sell_cushion_sd_window   | screen | sell_cushion_sd_window   | 20    |